Data Collection

```
import os
import pandas as pd

NFL_YEARS = [str(year) for year in range(2010, 2026)]

DATA_FILEPATH = r"Data"

def load_data(NFL_YEAR):
    path = os.path.join(DATA_FILEPATH, f"{NFL_YEAR}.csv")
    NFL = pd.read_csv(path)
    return NFL

for year in NFL_YEARS:
    df = load_data(year)
    print(f"{year} loaded — {len(df)} teams")

display(df.tail())
```

```
2010 loaded — 32 teams
2011 loaded — 32 teams
2012 loaded — 32 teams
2013 loaded — 32 teams
2014 loaded — 32 teams
2015 loaded — 32 teams
2016 loaded — 32 teams
2017 loaded — 32 teams
2018 loaded — 32 teams
2019 loaded — 32 teams
2020 loaded — 32 teams
2021 loaded — 32 teams
2022 loaded — 32 teams
2023 loaded — 32 teams
2024 loaded — 32 teams
2025 loaded — 32 teams
```

	Tm	W	L	T	W-L%	PF	PA	PD	MoV	SoS	SRS	OSRS	DSRS
27	New Orleans Saints	6	11	0	0.353	306	383	-77	-4.5	-0.3	-4.9	-5.5	0.6
28	Seattle Seahawks	14	3	0	0.824	483	292	191	11.2	1.6	12.8	6.2	6.7
29	Los Angeles	12	5	0	0.706	518	346	172	10.1	2.4	12.5	8.7	3.9

```
# 60/20/20 train/val/test split shared across all models
```

```
# Each model imports these globals instead of doing its own split

import os
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split

TEAM = 0; WINS = 1; LOSSES = 2; TIES = 3; WIN_PERCENTAGE = 4
POINTS_FOR = 5; POINTS_AGAINST = 6; POINTS_DIFFERENTIAL = 7
MARGIN_OF_VICTORY = 8; STRENGTH_OF_SCHEDULE = 9
SIMPLE_RATING_SYSTEM = 10; OFFENSIVE_SRS = 11; DEFENSIVE_SRS = 12

NFL_TEAMS = ['Arizona Cardinals', 'Atlanta Falcons', 'Baltimore Ravens
             'Carolina Panthers', 'Chicago Bears', 'Cincinnati Bengals
             'Dallas Cowboys', 'Denver Broncos', 'Detroit Lions', 'Gre
             'Houston Texans', 'Indianapolis Colts', 'Jacksonville Jag
             'Las Vegas Raiders', 'Los Angeles Chargers', 'Los Angeles
             'Minnesota Vikings', 'New England Patriots', 'New Orleans
             'New York Jets', 'Philadelphia Eagles', 'Pittsburgh Steel
             'Seattle Seahawks', 'Tampa Bay Buccaneers', 'Tennessee T
NFL_YEARS = [str(year) for year in range(2010, 2026)]
DATA_FILEPATH = "Data"
DROP_COLS = ['W', 'L', 'T', 'W-L%', 'PF', 'PA', 'MoV']


def _load_all_years():
    frames = []
    for year in NFL_YEARS:
        path = os.path.join(DATA_FILEPATH, f"{year}.csv")
        raw = pd.read_csv(path)
        raw = raw[raw.iloc[:, TEAM].isin(NFL_TEAMS)].copy()
        raw.columns = ['Team', 'W', 'L', 'T', 'W-L%', 'PF', 'PA', 'PD
                       'MoV', 'SoS', 'SRS', 'OSRS', 'DSRS']
        raw['Year'] = year
        frames.append(raw)
    combined = pd.concat(frames, ignore_index=True)
    for col in combined.columns:
        if col not in ('Team', 'Year'):
            combined[col] = pd.to_numeric(combined[col], errors='coer
    return combined.dropna()


_all_data = _load_all_years()
```

```
_features = _all_data[['SoS', 'SRS', 'OSRS', 'DSRS']]
_target = _all_data['W-L%']

# 60% train, 20% val, 20% test
X_train_global, _X_temp, y_train_global, _y_temp = train_test_split(
    _features.values, _target.values,
    test_size=0.40, random_state=42
)
X_val_global, X_test_global, y_val_global, y_test_global = train_test_
    _X_temp, _y_temp,
    test_size=0.50, random_state=42
)

print(f"Total samples : {len(_features)}")
print(f"Train (60%)   : {len(X_train_global)}")
print(f"Val   (20%)   : {len(X_val_global)}")
print(f"Test  (20%)   : {len(X_test_global)}")
```

```
Total samples : 512
Train (60%)   : 307
Val   (20%)   : 102
Test  (20%)   : 103
```

Logistic Regression Model

Greg Yonan

```
import os
import pickle
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

# Column indices
TEAM = 0; WINS = 1; LOSSES = 2; TIES = 3; WIN_PERCENTAGE = 4
POINTS_FOR = 5; POINTS_AGAINST = 6; POINTS_DIFFERENTIAL = 7
MARGIN_OF_VICTORY = 8; STRENGTH_OF_SCHEDULE = 9
SIMPLE_RATING_SYSTEM = 10; OFFENSIVE_SRS = 11; DEFENSIVE_SRS = 12

NFL_TEAMS = ['Arizona Cardinals', 'Atlanta Falcons', 'Baltimore Raven
             'Carolina Panthers', 'Chicago Bears', 'Cincinnati Bengal
             'Dallas Cowboys', 'Denver Broncos', 'Detroit Lions', 'Gr
             'Houston Texans', 'Indianapolis Colts', 'Jacksonville Ja
             'Las Vegas Raiders', 'Los Angeles Chargers', 'Los Angele
             'Minnesota Vikings', 'New England Patriots', 'New Orlean
             'New York Jets', 'Philadelphia Eagles', 'Pittsburgh Stee
             'Seattle Seahawks', 'Tampa Bay Buccaneers', 'Tennessee T
NFL_YEARS = [str(year) for year in range(2010, 2026)]

DATA_FILEPATH = "Data"
SCHEDULE_FILEPATH = "Schedule"
MODEL_FILEPATH = "Models/Logistic Regression"
PREDICTIONS_FILEPATH = "Predictions/Logistic Regression"
```

```
def read_data(NFL_YEAR, team_name):
    team_data = []
    with open(os.path.join(DATA_FILEPATH, f"{NFL_YEAR}.csv"), "r") as
        data = pd.read_csv(file)
        for index, row in data.iterrows():
            if row.iloc[TEAM] == team_name:
                team_data.append([
                    row.iloc[WINS], row.iloc[LOSSES], row.iloc[TIES],
                    row.iloc[POINTS_FOR], row.iloc[POINTS_AGAINST], r
                    row.iloc[MARGIN_OF_VICTORY], row.iloc[STRENGTH_OF_
                    row.iloc[SIMPLE_RATING_SYSTEM], row.iloc[OFFENSIV
                ])
    return pd.DataFrame(team_data, columns=['W', 'L', 'T', 'W-L%', 'P
```

```
def train_model():
    model = LinearRegression()
    model.fit(X_train_global, y_train_global)
    return model
```

```
def save_model(NFL_YEAR, model):
    os.makedirs(MODEL_FILEPATH, exist_ok=True)
    with open(os.path.join(MODEL_FILEPATH, f"linear_regression_model_
        pickle.dump(model, file)
```

```
def make_predictions(NFL_YEAR):
    model_path = os.path.join(MODEL_FILEPATH, "linear_regression_model_
    with open(model_path, "rb") as file:
        model = pickle.load(file)

    schedule_df = pd.read_csv(os.path.join(SCHEDULE_FILEPATH, f"{NFL_YE
    if schedule_df.empty:
        print(f"No schedule found for {NFL_YEAR}")
        return

    team_wins  = {team: 0 for team in NFL_TEAMS}
    team_games = {team: 0 for team in NFL_TEAMS}

    try:
        season_data = pd.read_csv(os.path.join(DATA_FILEPATH, f"{NFL_YE
        team_stats = {row.iloc[TEAM]: read_data(NFL_YEAR, row.iloc[TEAM
        team_stats = {k: v for k, v in team_stats.items() if v is not N
    except FileNotFoundError:
        print(f"Data file for {NFL_YEAR} not found")
        return

    drop_cols = ['W', 'L', 'T', 'W-L%', 'PF', 'PA', 'PD', 'MoV']
    for _, row in schedule_df.iterrows():
        winner, loser = row['Winner/tie'], row['Loser/tie']
        if winner not in team_stats or loser not in team_stats:
            continue
        winner_pred = model.predict(team_stats[winner].drop(columns=dro
        loser_pred  = model.predict(team_stats[loser].drop(columns=drop
        predicted_winner = winner if winner_pred > loser_pred else lose
        predicted_loser  = loser  if winner_pred > loser_pred else winn
        team_wins[predicted_winner] += 1
        team_games[predicted_winner] += 1
```

```
            team_games[predicted_loser]  += 1

        predictions = [
            [team,
             team_stats[team]['W-L%'].iloc[0] if team in team_stats else 0.
             team_wins[team] / team_games[team] if team_games[team] > 0 els
            for team in NFL_TEAMS
        ]
        predictions_df = pd.DataFrame(predictions, columns=['Team', 'Actual
        os.makedirs(PREDICTIONS_FILEPATH, exist_ok=True)
        predictions_df.to_csv(os.path.join(PREDICTIONS_FILEPATH, f"predicti
        return predictions_df
```

```
model = train_model()
save_model("global", model)
print("Model saved.")

# Evaluate on the global validation split
from sklearn.metrics import r2_score, mean_squared_error
y_pred_lr_val = model.predict(X_val_global)
print(f"R²: {r2_score(y_val_global, y_pred_lr_val):.4f}")
print(f"MSE: {mean_squared_error(y_val_global, y_pred_lr_val):.6f}")

# Run game-by-game predictions for each year using the global model
for year in NFL_YEARS:
    print(f"\nPredictions for {year}...")
    preds = make_predictions(year)
    if preds is not None:
        display(preds)
```

```
Model saved.
R²: 0.8426
MSE: 0.006066

Predictions for 2010...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.313	0.0625
1	Atlanta Falcons	0.813	0.8750
2	Baltimore Ravens	0.750	0.7500
3	Buffalo Bills	0.250	0.0000
4	Carolina Panthers	0.125	0.0000

4	Carolina Panthers	[illegible]	[illegible]
5	Chicago Bears	0.688	0.6875
6	Cincinnati Bengals	0.250	0.1250
7	Cleveland Browns	0.313	0.3750
8	Dallas Cowboys	0.375	0.3125
9	Denver Broncos	0.250	0.0625
10	Detroit Lions	0.375	0.4375
11	Green Bay Packers	0.625	0.9375
12	Houston Texans	0.375	0.3125
13	Indianapolis Colts	0.625	0.6875
14	Jacksonville Jaguars	0.500	0.1875
15	Kansas City Chiefs	0.625	0.8750
16	Las Vegas Raiders	0.500	0.6250
17	Los Angeles Chargers	0.563	0.9375
18	Los Angeles Rams	0.438	0.5625
19	Miami Dolphins	0.438	0.2500
20	Minnesota Vikings	0.375	0.1875
21	New England Patriots	0.875	1.0000
22	New Orleans Saints	0.688	0.7500
23	New York Giants	0.625	0.7500
24	New York Jets	0.688	0.6875
25	Philadelphia Eagles	0.625	0.8750
26	Pittsburgh Steelers	0.750	0.9375
27	San Francisco 49ers	0.375	0.3750
28	Seattle Seahawks	0.438	0.2500
29	Tampa Bay Buccaneers	0.625	0.6250
30	Tennessee Titans	0.375	0.5000
31	Washington Commanders	0.375	0.0000

```
Predictions for 2011...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.500	0.3125
1	Atlanta Falcons	0.625	0.6250
2	Baltimore Ravens	0.750	0.9375
3	Buffalo Bills	0.375	0.2500
4	Carolina Panthers	0.375	0.4375
5	Chicago Bears	0.500	0.5000
6	Cincinnati Bengals	0.563	0.6250
7	Cleveland Browns	0.250	0.1250
8	Dallas Cowboys	0.500	0.6250
9	Denver Broncos	0.500	0.1875
10	Detroit Lions	0.625	0.7500
11	Green Bay Packers	0.938	0.9375
12	Houston Texans	0.625	0.8750
13	Indianapolis Colts	0.125	0.0625
14	Jacksonville Jaguars	0.313	0.2500
15	Kansas City Chiefs	0.438	0.0625
16	Las Vegas Raiders	0.500	0.3750
17	Los Angeles Chargers	0.500	0.7500
18	Los Angeles Rams	0.125	0.0000
19	Miami Dolphins	0.375	0.6875
20	Minnesota Vikings	0.188	0.1250
21	New England Patriots	0.813	1.0000
22	New Orleans Saints	0.813	1.0000
23	New York Giants	0.563	0.3125
24	New York Jets	0.500	0.5000
25	Philadelphia Eagles	0.500	0.8750

26	Pittsburgh Steelers	0.750	0.6875
27	San Francisco 49ers	0.813	1.0000
28	Seattle Seahawks	0.438	0.4375
29	Tampa Bay Buccaneers	0.250	0.0000
30	Tennessee Titans	0.563	0.5625
31	Washington Commanders	0.313	0.1250

```
Predictions for 2012...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.313	0.0625
1	Atlanta Falcons	0.813	0.9375
2	Baltimore Ravens	0.625	0.6250
3	Buffalo Bills	0.375	0.3750
4	Carolina Panthers	0.438	0.2500
5	Chicago Bears	0.625	0.8750
6	Cincinnati Bengals	0.625	0.8750
7	Cleveland Browns	0.313	0.2500
8	Dallas Cowboys	0.500	0.1875
9	Denver Broncos	0.813	0.9375
10	Detroit Lions	0.250	0.2500
11	Green Bay Packers	0.688	0.7500
12	Houston Texans	0.750	0.7500
13	Indianapolis Colts	0.688	0.5625
14	Jacksonville Jaguars	0.125	0.0000
15	Kansas City Chiefs	0.125	0.0000
16	Las Vegas Raiders	0.250	0.1875
17	Los Angeles Chargers	0.438	0.5625
18	Los Angeles Rams	0.469	0.3125

19	Miami Dolphins	0.438	0.6250
20	Minnesota Vikings	0.625	0.5000
21	New England Patriots	0.750	1.0000
22	New Orleans Saints	0.438	0.5625
23	New York Giants	0.563	0.8125
24	New York Jets	0.375	0.1875
25	Philadelphia Eagles	0.250	0.0000
26	Pittsburgh Steelers	0.500	0.5625
27	San Francisco 49ers	0.719	0.8125
28	Seattle Seahawks	0.688	0.9375
29	Tampa Bay Buccaneers	0.438	0.4375
30	Tennessee Titans	0.375	0.1250
31	Washington Commanders	0.625	0.6875

```
Predictions for 2013...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.625	0.5000
1	Atlanta Falcons	0.250	0.2500
2	Baltimore Ravens	0.500	0.3750
3	Buffalo Bills	0.375	0.3750
4	Carolina Panthers	0.750	0.8750
5	Chicago Bears	0.500	0.3750
6	Cincinnati Bengals	0.688	1.0000
7	Cleveland Browns	0.250	0.1250
8	Dallas Cowboys	0.500	0.5625
9	Denver Broncos	0.813	1.0000
10	Detroit Lions	0.438	0.8125
11	Green Bay Packers	0.531	0.5625
12	Houston Texans	0.125	0.1250

13	Indianapolis Colts	0.688	0.6875
14	Jacksonville Jaguars	0.250	0.0000
15	Kansas City Chiefs	0.688	0.8750
16	Las Vegas Raiders	0.250	0.1875
17	Los Angeles Chargers	0.563	0.5625
18	Los Angeles Rams	0.438	0.3750
19	Miami Dolphins	0.500	0.5000
20	Minnesota Vikings	0.344	0.1250
21	New England Patriots	0.750	0.7500
22	New Orleans Saints	0.688	0.7500
23	New York Giants	0.438	0.2500
24	New York Jets	0.500	0.1250
25	Philadelphia Eagles	0.625	0.8750
26	Pittsburgh Steelers	0.500	0.7500
27	San Francisco 49ers	0.750	0.8750
28	Seattle Seahawks	0.813	1.0000
29	Tampa Bay Buccaneers	0.250	0.0000
30	Tennessee Titans	0.438	0.3750
31	Washington Commanders	0.188	0.0000

```
Predictions for 2014...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.688	0.5625
1	Atlanta Falcons	0.375	0.2500
2	Baltimore Ravens	0.625	1.0000
3	Buffalo Bills	0.563	0.6250
4	Carolina Panthers	0.469	0.3750
5	Chicago Bears	0.313	0.0625

6	Cincinnati Bengals	0.656	0.5000
7	Cleveland Browns	0.438	0.2500
8	Dallas Cowboys	0.750	0.9375
9	Denver Broncos	0.750	0.8750
10	Detroit Lions	0.688	0.7500
11	Green Bay Packers	0.750	0.8750
12	Houston Texans	0.563	0.6250
13	Indianapolis Colts	0.688	0.7500
14	Jacksonville Jaguars	0.188	0.1250
15	Kansas City Chiefs	0.563	0.6875
16	Las Vegas Raiders	0.188	0.0000
17	Los Angeles Chargers	0.563	0.3750
18	Los Angeles Rams	0.375	0.3125
19	Miami Dolphins	0.500	0.4375
20	Minnesota Vikings	0.438	0.5625
21	New England Patriots	0.750	1.0000
22	New Orleans Saints	0.438	0.5625
23	New York Giants	0.375	0.4375
24	New York Jets	0.250	0.1875
25	Philadelphia Eagles	0.625	0.6875
26	Pittsburgh Steelers	0.688	0.8125
27	San Francisco 49ers	0.500	0.1875
28	Seattle Seahawks	0.750	1.0000
29	Tampa Bay Buccaneers	0.125	0.0625
30	Tennessee Titans	0.125	0.0000
31	Washington Commanders	0.250	0.1250

```
Predictions for 2015...
```

Team	Actual W-L%	Predicted W-L%

0	Arizona Cardinals	0.813	1.0000
1	Atlanta Falcons	0.500	0.6875
2	Baltimore Ravens	0.313	0.3750
3	Buffalo Bills	0.500	0.5625
4	Carolina Panthers	0.938	1.0000
5	Chicago Bears	0.375	0.1875
6	Cincinnati Bengals	0.750	0.8750
7	Cleveland Browns	0.188	0.0000
8	Dallas Cowboys	0.250	0.0000
9	Denver Broncos	0.750	0.6250
10	Detroit Lions	0.438	0.4375
11	Green Bay Packers	0.625	0.5625
12	Houston Texans	0.563	0.6875
13	Indianapolis Colts	0.500	0.3125
14	Jacksonville Jaguars	0.313	0.2500
15	Kansas City Chiefs	0.688	0.9375
16	Las Vegas Raiders	0.438	0.4375
17	Los Angeles Chargers	0.250	0.1250
18	Los Angeles Rams	0.438	0.3750
19	Miami Dolphins	0.375	0.1250
20	Minnesota Vikings	0.688	0.8125
21	New England Patriots	0.750	1.0000
22	New Orleans Saints	0.438	0.3750
23	New York Giants	0.375	0.5000
24	New York Jets	0.625	0.8750
25	Philadelphia Eagles	0.438	0.3125
26	Pittsburgh Steelers	0.625	0.6250
27	San Francisco 49ers	0.313	0.0625

27	San Francisco 49ers	0.313	0.0625
28	Seattle Seahawks	0.625	0.8125
29	Tampa Bay Buccaneers	0.375	0.2500
30	Tennessee Titans	0.188	0.0625
31	Washington Commanders	0.563	0.7500

```
Predictions for 2016...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.469	0.7500
1	Atlanta Falcons	0.688	1.0000
2	Baltimore Ravens	0.500	0.5625
3	Buffalo Bills	0.438	0.5625
4	Carolina Panthers	0.375	0.1250
5	Chicago Bears	0.188	0.0625
6	Cincinnati Bengals	0.406	0.3125
7	Cleveland Browns	0.063	0.0000
8	Dallas Cowboys	0.813	1.0000
9	Denver Broncos	0.563	0.7500
10	Detroit Lions	0.563	0.3125
11	Green Bay Packers	0.625	0.8125
12	Houston Texans	0.563	0.1875
13	Indianapolis Colts	0.500	0.6250
14	Jacksonville Jaguars	0.188	0.0625
15	Kansas City Chiefs	0.750	0.9375
16	Las Vegas Raiders	0.750	0.6875
17	Los Angeles Chargers	0.313	0.2500
18	Los Angeles Rams	0.250	0.1250
19	Miami Dolphins	0.625	0.3750
20	Minnesota Vikings	0.500	0.6250

21	New England Patriots	0.875	1.0000
22	New Orleans Saints	0.438	0.5000
23	New York Giants	0.688	0.6250
24	New York Jets	0.313	0.1875
25	Philadelphia Eagles	0.438	0.6250
26	Pittsburgh Steelers	0.688	0.8125
27	San Francisco 49ers	0.125	0.0000
28	Seattle Seahawks	0.656	0.8750
29	Tampa Bay Buccaneers	0.563	0.3750
30	Tennessee Titans	0.563	0.5625
31	Washington Commanders	0.531	0.3125

```
Predictions for 2017...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.500	0.1875
1	Atlanta Falcons	0.625	0.6250
2	Baltimore Ravens	0.563	0.7500
3	Buffalo Bills	0.563	0.5000
4	Carolina Panthers	0.688	0.6875
5	Chicago Bears	0.313	0.2500
6	Cincinnati Bengals	0.438	0.3750
7	Cleveland Browns	0.000	0.0000
8	Dallas Cowboys	0.563	0.5625
9	Denver Broncos	0.313	0.1875
10	Detroit Lions	0.563	0.5625
11	Green Bay Packers	0.438	0.0625
12	Houston Texans	0.250	0.1875
13	Indianapolis Colts	0.250	0.0625
14	Jacksonville Jaguars	0.625	1.0000

14	Jacksonville Jaguars	0.625	1.0000
15	Kansas City Chiefs	0.625	0.6875
16	Las Vegas Raiders	0.375	0.3125
17	Los Angeles Chargers	0.563	0.8125
18	Los Angeles Rams	0.688	0.8750
19	Miami Dolphins	0.375	0.0000
20	Minnesota Vikings	0.813	0.9375
21	New England Patriots	0.813	1.0000
22	New Orleans Saints	0.688	0.8125
23	New York Giants	0.188	0.0000
24	New York Jets	0.313	0.2500
25	Philadelphia Eagles	0.813	1.0000
26	Pittsburgh Steelers	0.813	0.8125
27	San Francisco 49ers	0.375	0.3750
28	Seattle Seahawks	0.563	0.6875
29	Tampa Bay Buccaneers	0.313	0.4375
30	Tennessee Titans	0.563	0.6250
31	Washington Commanders	0.438	0.3750

```
Predictions for 2018...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.188	0.0000
1	Atlanta Falcons	0.438	0.5000
2	Baltimore Ravens	0.625	0.8750
3	Buffalo Bills	0.375	0.2500
4	Carolina Panthers	0.438	0.5625
5	Chicago Bears	0.750	1.0000
6	Cincinnati Bengals	0.375	0.1250
7	Cleveland Browns	0.469	0.3125

8	Dallas Cowboys	0.625	0.5625
9	Denver Broncos	0.375	0.4375
10	Detroit Lions	0.375	0.3125
11	Green Bay Packers	0.406	0.5000
12	Houston Texans	0.688	0.8125
13	Indianapolis Colts	0.625	0.9375
14	Jacksonville Jaguars	0.313	0.2500
15	Kansas City Chiefs	0.750	1.0000
16	Las Vegas Raiders	0.250	0.0625
17	Los Angeles Chargers	0.750	0.7500
18	Los Angeles Rams	0.813	0.8125
19	Miami Dolphins	0.438	0.0625
20	Minnesota Vikings	0.531	0.5625
21	New England Patriots	0.688	0.8750
22	New Orleans Saints	0.813	1.0000
23	New York Giants	0.313	0.3125
24	New York Jets	0.250	0.1250
25	Philadelphia Eagles	0.563	0.7500
26	Pittsburgh Steelers	0.594	0.6250
27	San Francisco 49ers	0.250	0.1875
28	Seattle Seahawks	0.625	0.6875
29	Tampa Bay Buccaneers	0.313	0.1875
30	Tennessee Titans	0.563	0.5000
31	Washington Commanders	0.438	0.0625

```
Predictions for 2019...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.344	0.1875
1	Atlanta Falcons	0.438	0.2500

1	Atlanta Falcons	0.438	0.2500
2	Baltimore Ravens	0.875	1.0000
3	Buffalo Bills	0.625	0.6875
4	Carolina Panthers	0.313	0.0625
5	Chicago Bears	0.500	0.3750
6	Cincinnati Bengals	0.125	0.0625
7	Cleveland Browns	0.375	0.3125
8	Dallas Cowboys	0.500	0.9375
9	Denver Broncos	0.438	0.3125
10	Detroit Lions	0.219	0.2500
11	Green Bay Packers	0.813	0.6875
12	Houston Texans	0.625	0.5000
13	Indianapolis Colts	0.438	0.5000
14	Jacksonville Jaguars	0.375	0.1875
15	Kansas City Chiefs	0.750	0.8750
16	Las Vegas Raiders	0.438	0.0625
17	Los Angeles Chargers	0.313	0.6875
18	Los Angeles Rams	0.563	0.6875
19	Miami Dolphins	0.313	0.0000
20	Minnesota Vikings	0.625	0.8750
21	New England Patriots	0.750	0.9375
22	New Orleans Saints	0.813	0.8750
23	New York Giants	0.250	0.1875
24	New York Jets	0.438	0.4375
25	Philadelphia Eagles	0.563	0.6250
26	Pittsburgh Steelers	0.500	0.4375
27	San Francisco 49ers	0.813	0.9375
28	Seattle Seahawks	0.688	0.4375

29	Tampa Bay Buccaneers	0.438	0.6875
30	Tennessee Titans	0.563	0.8750
31	Washington Commanders	0.188	0.0625

```
Predictions for 2020...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.500	0.6250
1	Atlanta Falcons	0.250	0.5000
2	Baltimore Ravens	0.688	1.0000
3	Buffalo Bills	0.813	1.0000
4	Carolina Panthers	0.313	0.1250
5	Chicago Bears	0.500	0.5625
6	Cincinnati Bengals	0.281	0.0625
7	Cleveland Browns	0.688	0.5625
8	Dallas Cowboys	0.375	0.1875
9	Denver Broncos	0.313	0.0625
10	Detroit Lions	0.313	0.0625
11	Green Bay Packers	0.813	0.9375
12	Houston Texans	0.250	0.2500
13	Indianapolis Colts	0.688	0.8125
14	Jacksonville Jaguars	0.063	0.0000
15	Kansas City Chiefs	0.875	0.7500
16	Las Vegas Raiders	0.500	0.2500
17	Los Angeles Chargers	0.438	0.5000
18	Los Angeles Rams	0.625	0.7500
19	Miami Dolphins	0.625	0.6875
20	Minnesota Vikings	0.438	0.3750
21	New England Patriots	0.438	0.3750
22	New Orleans Saints	0.750	1.0000

23	New York Giants	0.375	0.3125
24	New York Jets	0.125	0.0000
25	Philadelphia Eagles	0.281	0.0625
26	Pittsburgh Steelers	0.750	0.8125
27	San Francisco 49ers	0.375	0.3125
28	Seattle Seahawks	0.750	0.9375
29	Tampa Bay Buccaneers	0.688	0.8125
30	Tennessee Titans	0.688	0.6250
31	Washington Commanders	0.438	0.6875

```
Predictions for 2021...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.647	0.705882
1	Atlanta Falcons	0.412	0.176471
2	Baltimore Ravens	0.471	0.470588
3	Buffalo Bills	0.647	1.000000
4	Carolina Panthers	0.294	0.294118
5	Chicago Bears	0.353	0.176471
6	Cincinnati Bengals	0.588	0.941176
7	Cleveland Browns	0.471	0.352941
8	Dallas Cowboys	0.706	1.000000
9	Denver Broncos	0.412	0.705882
10	Detroit Lions	0.206	0.058824
11	Green Bay Packers	0.765	0.823529
12	Houston Texans	0.235	0.176471
13	Indianapolis Colts	0.529	0.764706
14	Jacksonville Jaguars	0.176	0.000000
15	Kansas City Chiefs	0.706	0.882353

16	Las Vegas Raiders	0.588	0.176471
17	Los Angeles Chargers	0.529	0.529412
18	Los Angeles Rams	0.706	0.941176
19	Miami Dolphins	0.529	0.470588
20	Minnesota Vikings	0.471	0.470588
21	New England Patriots	0.588	0.764706
22	New Orleans Saints	0.529	0.529412
23	New York Giants	0.235	0.000000
24	New York Jets	0.235	0.058824
25	Philadelphia Eagles	0.529	0.764706
26	Pittsburgh Steelers	0.559	0.176471
27	San Francisco 49ers	0.588	0.470588
28	Seattle Seahawks	0.412	0.411765
29	Tampa Bay Buccaneers	0.765	0.882353
30	Tennessee Titans	0.706	0.588235
31	Washington Commanders	0.412	0.235294

```
Predictions for 2022...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.235	0.000000
1	Atlanta Falcons	0.412	0.588235
2	Baltimore Ravens	0.588	0.764706
3	Buffalo Bills	0.813	1.000000
4	Carolina Panthers	0.412	0.411765
5	Chicago Bears	0.176	0.000000
6	Cincinnati Bengals	0.750	0.875000
7	Cleveland Browns	0.412	0.235294
8	Dallas Cowboys	0.706	0.882353
9	Denver Broncos	0.294	0.235294

10	Detroit Lions	0.529	0.764706
11	Green Bay Packers	0.471	0.647059
12	Houston Texans	0.206	0.176471
13	Indianapolis Colts	0.265	0.000000
14	Jacksonville Jaguars	0.529	0.823529
15	Kansas City Chiefs	0.824	0.882353
16	Las Vegas Raiders	0.353	0.470588
17	Los Angeles Chargers	0.588	0.764706
18	Los Angeles Rams	0.294	0.117647
19	Miami Dolphins	0.529	0.352941
20	Minnesota Vikings	0.765	0.529412
21	New England Patriots	0.471	0.705882
22	New Orleans Saints	0.412	0.647059
23	New York Giants	0.559	0.411765
24	New York Jets	0.412	0.235294
25	Philadelphia Eagles	0.824	1.000000
26	Pittsburgh Steelers	0.529	0.117647
27	San Francisco 49ers	0.765	1.000000
28	Seattle Seahawks	0.529	0.705882
29	Tampa Bay Buccaneers	0.471	0.117647
30	Tennessee Titans	0.412	0.294118
31	Washington Commanders	0.500	0.294118

```
Predictions for 2023...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.235	0.117647
1	Atlanta Falcons	0.412	0.352941
2	Baltimore Ravens	0.765	1.000000

3	Buffalo Bills	0.647	0.941176
4	Carolina Panthers	0.118	0.000000
5	Chicago Bears	0.412	0.352941
6	Cincinnati Bengals	0.529	0.235294
7	Cleveland Browns	0.647	0.823529
8	Dallas Cowboys	0.706	1.000000
9	Denver Broncos	0.471	0.176471
10	Detroit Lions	0.706	0.764706
11	Green Bay Packers	0.529	0.764706
12	Houston Texans	0.588	0.823529
13	Indianapolis Colts	0.529	0.294118
14	Jacksonville Jaguars	0.529	0.470588
15	Kansas City Chiefs	0.647	0.882353
16	Las Vegas Raiders	0.471	0.647059
17	Los Angeles Chargers	0.294	0.294118
18	Los Angeles Rams	0.588	0.588235
19	Miami Dolphins	0.647	0.764706
20	Minnesota Vikings	0.412	0.411765
21	New England Patriots	0.235	0.117647
22	New Orleans Saints	0.529	1.000000
23	New York Giants	0.353	0.117647
24	New York Jets	0.412	0.235294
25	Philadelphia Eagles	0.647	0.529412
26	Pittsburgh Steelers	0.588	0.411765
27	San Francisco 49ers	0.706	0.882353
28	Seattle Seahawks	0.529	0.352941
29	Tampa Bay Buccaneers	0.529	0.588235
30	Tennessee Titans	0.353	0.058824

31	Washington Commanders	0.235	0.000000

```
Predictions for 2024...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.471	0.647059
1	Atlanta Falcons	0.471	0.411765
2	Baltimore Ravens	0.706	0.941176
3	Buffalo Bills	0.765	0.882353
4	Carolina Panthers	0.294	0.000000
5	Chicago Bears	0.294	0.235294
6	Cincinnati Bengals	0.529	0.588235
7	Cleveland Browns	0.176	0.000000
8	Dallas Cowboys	0.412	0.235294
9	Denver Broncos	0.588	0.882353
10	Detroit Lions	0.882	1.000000
11	Green Bay Packers	0.647	0.823529
12	Houston Texans	0.588	0.647059
13	Indianapolis Colts	0.471	0.470588
14	Jacksonville Jaguars	0.235	0.294118
15	Kansas City Chiefs	0.882	0.588235
16	Las Vegas Raiders	0.235	0.117647
17	Los Angeles Chargers	0.647	0.764706
18	Los Angeles Rams	0.588	0.470588
19	Miami Dolphins	0.471	0.588235
20	Minnesota Vikings	0.824	0.764706
21	New England Patriots	0.235	0.058824
22	New Orleans Saints	0.294	0.352941
23	New York Giants	0.176	0.117647

24	New York Jets	0.294	0.235294
25	Philadelphia Eagles	0.824	1.000000
26	Pittsburgh Steelers	0.588	0.470588
27	San Francisco 49ers	0.353	0.235294
28	Seattle Seahawks	0.588	0.588235
29	Tampa Bay Buccaneers	0.588	0.823529
30	Tennessee Titans	0.176	0.000000
31	Washington Commanders	0.706	0.764706

```
Predictions for 2025...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.176	0.058824
1	Atlanta Falcons	0.471	0.470588
2	Baltimore Ravens	0.471	0.529412
3	Buffalo Bills	0.706	0.882353
4	Carolina Panthers	0.471	0.294118
5	Chicago Bears	0.647	0.764706
6	Cincinnati Bengals	0.353	0.235294
7	Cleveland Browns	0.294	0.176471
8	Dallas Cowboys	0.441	0.470588
9	Denver Broncos	0.824	0.882353
10	Detroit Lions	0.529	0.941176
11	Green Bay Packers	0.559	0.647059
12	Houston Texans	0.706	0.705882
13	Indianapolis Colts	0.471	0.529412
14	Jacksonville Jaguars	0.765	0.882353
15	Kansas City Chiefs	0.353	0.529412
16	Las Vegas Raiders	0.176	0.058824
17	Los Angeles Chargers	0.647	0.529412

18	Los Angeles Rams	0.706	0.882353
19	Miami Dolphins	0.412	0.352941
20	Minnesota Vikings	0.529	0.411765
21	New England Patriots	0.824	1.000000
22	New Orleans Saints	0.353	0.176471
23	New York Giants	0.235	0.235294
24	New York Jets	0.176	0.000000
25	Philadelphia Eagles	0.647	0.764706
26	Pittsburgh Steelers	0.588	0.352941
27	San Francisco 49ers	0.706	0.647059
28	Seattle Seahawks	0.824	1.000000
29	Tampa Bay Buccaneers	0.471	0.529412
30	Tennessee Titans	0.176	0.000000
31	Washington Commanders	0.294	0.058824

Next steps: Generate code with `preds` New interactive sheet Generate code with `preds`

Neural Network Model

Dylan Shaffer

```
import os
import pickle
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from tensorflow import keras
from tensorflow.keras import layers

TEAM = 0; WINS = 1; LOSSES = 2; TIES = 3; WIN_PERCENTAGE = 4
POINTS_FOR = 5; POINTS_AGAINST = 6; POINTS_DIFFERENTIAL = 7
MARGIN_OF_VICTORY = 8; STRENGTH_OF_SCHEDULE = 9
SIMPLE_RATING_SYSTEM = 10; OFFENSIVE_SRS = 11; DEFENSIVE_SRS = 12

DATA_FILEPATH = "Data"
MODEL_FILEPATH = "Models/Neural Network"
PREDICTIONS_FILEPATH = "Predictions/Neural Network"

DROP_COLS = ['W', 'L', 'T', 'W-L%', 'PF', 'PA', 'PD', 'MoV']
```

```
def read_data(NFL_YEAR, team_name):
    team_data = []
    with open(os.path.join(DATA_FILEPATH, f"{NFL_YEAR}.csv"), "r") as
        data = pd.read_csv(file)
        for index, row in data.iterrows():
            if row.iloc[TEAM] == team_name:
                team_data.append([
                    row.iloc[WINS], row.iloc[LOSSES], row.iloc[TIES],
                    row.iloc[POINTS_FOR], row.iloc[POINTS_AGAINST], r
                    row.iloc[MARGIN_OF_VICTORY], row.iloc[STRENGTH_OF_
                    row.iloc[SIMPLE_RATING_SYSTEM], row.iloc[OFFENSIVI
                ])
    return pd.DataFrame(team_data, columns=['W', 'L', 'T', 'W-L%', 'PI
```

```
def build_model(input_dim):
    model = keras.Sequential([
        layers.Dense(64, activation="relu", input_shape=(input_dim,)),
        layers.Dense(64, activation="relu"),
        layers.Dense(1)  # single output: W-L%
    ])
    model.compile(optimizer="rmsprop", loss="mse", metrics=["mae"])
    return model
```

```
def train_model():
    mean = X_train_global.mean(axis=0)
    std = X_train_global.std(axis=0)

    X_train_norm = (X_train_global - mean) / std
    X_val_norm = (X_val_global - mean) / std

    model = build_model(X_train_norm.shape[1])

    early_stopping = keras.callbacks.EarlyStopping(
        monitor="val_loss",
        patience=10,
        restore_best_weights=True,
        verbose=1
    )

    # log every 10 epochs so we can see where val_loss plateaus
    epoch_logger = keras.callbacks.LambdaCallback(
        on_epoch_end=lambda epoch, logs: print(
            f"  Epoch {epoch+1:3d} - loss: {logs['loss']:.6f}  val_los
        ) if (epoch + 1) % 10 == 0 else None
    )

    print("Training (max 100 epochs, early stopping patience=10)...")
    history = model.fit(
        X_train_norm, y_train_global,
        epochs=100, batch_size=8, verbose=0,
        validation_data=(X_val_norm, y_val_global),
        callbacks=[early_stopping, epoch_logger]
    )

    stopped_epoch = early_stopping.stopped_epoch
    best_epoch = stopped_epoch - early_stopping.patience + 1 if stoppe
    best_val_loss = min(history.history["val_loss"])
    print(f"\nDone. Best epoch: {best_epoch}, best val_loss: {best_va

    return model, mean, std
```

```
def save_model(NFL_YEAR, model, mean, std):
    os.makedirs(MODEL_FILEPATH, exist_ok=True)
    model.save(os.path.join(MODEL_FILEPATH, f"nn_model_{NFL_YEAR}.kera
    np.save(os.path.join(MODEL_FILEPATH, f"nn_mean_{NFL_YEAR}.npy"), r
    np.save(os.path.join(MODEL_FILEPATH, f"nn_std_{NFL_YEAR}.npy"),  s
```

```
def make_predictions(NFL_YEAR):
    model = keras.models.load_model(os.path.join(MODEL_FILEPATH, "nn_mo
    mean = np.load(os.path.join(MODEL_FILEPATH, "nn_mean_global.npy"))
    std = np.load(os.path.join(MODEL_FILEPATH, "nn_std_global.npy"))

    schedule_df = pd.read_csv(os.path.join("Schedule", f"{NFL_YEAR}.csv
    if schedule_df.empty:
        print(f"No schedule found for {NFL_YEAR}")
        return

    season_data = pd.read_csv(os.path.join(DATA_FILEPATH, f"{NFL_YEAR}.
    team_stats  = {row.iloc[TEAM]: read_data(NFL_YEAR, row.iloc[TEAM])
    team_stats  = {k: v for k, v in team_stats.items() if v is not None

    team_wins  = {team: 0 for team in NFL_TEAMS}
    team_games = {team: 0 for team in NFL_TEAMS}

    for _, row in schedule_df.iterrows():
        winner, loser = row['Winner/tie'], row['Loser/tie']
        if winner not in team_stats or loser not in team_stats:
            continue

        def predict_team(team):
            X = team_stats[team].drop(columns=DROP_COLS).values
            X = (X - mean) / std
            return model.predict(X, verbose=0)[0][0]

        winner_pred = predict_team(winner)
        loser_pred  = predict_team(loser)

        predicted_winner = winner if winner_pred > loser_pred else lose
        predicted_loser  = loser  if winner_pred > loser_pred else winn
        team_wins[predicted_winner]  += 1
        team_games[predicted_winner] += 1
        team_games[predicted_loser]  += 1

    predictions = [
        [team,
```

```
            team_stats[team]['W-L%'].iloc[0] if team in team_stats else 0.
            team_wins[team] / team_games[team] if team_games[team] > 0 els
        for team in NFL_TEAMS
    ]
    predictions_df = pd.DataFrame(predictions, columns=['Team', 'Actual
    os.makedirs(PREDICTIONS_FILEPATH, exist_ok=True)
    predictions_df.to_csv(os.path.join(PREDICTIONS_FILEPATH, f"predicti
    return predictions_df
```

```
model, mean, std = train_model()
save_model("global", model, mean, std)
print("Model saved.")

# Run game-by-game predictions for each year using the global model
for year in NFL_YEARS:
    print(f"\nPredictions for {year}...")
    preds = make_predictions(year)
    if preds is not None:
        display(preds)
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Training (max 100 epochs, early stopping patience=10)...
  Epoch  10 — loss: 0.007075  val_loss: 0.009467  val_mae: 0.077503
Epoch 14: early stopping
Restoring model weights from the end of the best epoch: 4.

Done. Best epoch: 4, best val_loss: 0.006125, total epochs: 14
Model saved.

Predictions for 2010...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.313	0.1250
1	Atlanta Falcons	0.813	0.8750
2	Baltimore Ravens	0.750	0.6875
3	Buffalo Bills	0.250	0.0000
4	Carolina Panthers	0.125	0.0000
5	Chicago Bears	0.688	0.6875
6	Cincinnati Bengals	0.250	0.1250

7	Cleveland Browns	0.313	0.3125
8	Dallas Cowboys	0.375	0.3125
9	Denver Broncos	0.250	0.0000
10	Detroit Lions	0.375	0.5000
11	Green Bay Packers	0.625	0.9375
12	Houston Texans	0.375	0.3125
13	Indianapolis Colts	0.625	0.7500
14	Jacksonville Jaguars	0.500	0.1875
15	Kansas City Chiefs	0.625	0.6875
16	Las Vegas Raiders	0.500	0.8125
17	Los Angeles Chargers	0.563	0.9375
18	Los Angeles Rams	0.438	0.5625
19	Miami Dolphins	0.438	0.3125
20	Minnesota Vikings	0.375	0.1875
21	New England Patriots	0.875	1.0000
22	New Orleans Saints	0.688	0.7500
23	New York Giants	0.625	0.6875
24	New York Jets	0.688	0.7500
25	Philadelphia Eagles	0.625	0.8750
26	Pittsburgh Steelers	0.750	0.9375
27	San Francisco 49ers	0.375	0.3750
28	Seattle Seahawks	0.438	0.2500
29	Tampa Bay Buccaneers	0.625	0.5625
30	Tennessee Titans	0.375	0.5000
31	Washington Commanders	0.375	0.0000

```
Predictions for 2011...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.500	0.3750

1	Atlanta Falcons	0.625	0.6250
2	Baltimore Ravens	0.750	0.8125
3	Buffalo Bills	0.375	0.2500
4	Carolina Panthers	0.375	0.3750
5	Chicago Bears	0.500	0.5000
6	Cincinnati Bengals	0.563	0.6250
7	Cleveland Browns	0.250	0.1875
8	Dallas Cowboys	0.500	0.5625
9	Denver Broncos	0.500	0.3125
10	Detroit Lions	0.625	0.7500
11	Green Bay Packers	0.938	1.0000
12	Houston Texans	0.625	0.8125
13	Indianapolis Colts	0.125	0.0000
14	Jacksonville Jaguars	0.313	0.1875
15	Kansas City Chiefs	0.438	0.0625
16	Las Vegas Raiders	0.500	0.2500
17	Los Angeles Chargers	0.500	0.7500
18	Los Angeles Rams	0.125	0.0000
19	Miami Dolphins	0.375	0.5625
20	Minnesota Vikings	0.188	0.1250
21	New England Patriots	0.813	1.0000
22	New Orleans Saints	0.813	0.9375
23	New York Giants	0.563	0.3125
24	New York Jets	0.500	0.6875
25	Philadelphia Eagles	0.500	0.8750
26	Pittsburgh Steelers	0.750	0.8750
27	San Francisco 49ers	0.813	1.0000

28	Seattle Seahawks	0.438	0.4375
29	Tampa Bay Buccaneers	0.250	0.0625
30	Tennessee Titans	0.563	0.5625
31	Washington Commanders	0.313	0.1250

```
Predictions for 2012...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.313	0.0625
1	Atlanta Falcons	0.813	0.9375
2	Baltimore Ravens	0.625	0.6875
3	Buffalo Bills	0.375	0.2500
4	Carolina Panthers	0.438	0.3750
5	Chicago Bears	0.625	0.8750
6	Cincinnati Bengals	0.625	0.6875
7	Cleveland Browns	0.313	0.2500
8	Dallas Cowboys	0.500	0.2500
9	Denver Broncos	0.813	0.9375
10	Detroit Lions	0.250	0.2500
11	Green Bay Packers	0.688	0.7500
12	Houston Texans	0.750	0.7500
13	Indianapolis Colts	0.688	0.6250
14	Jacksonville Jaguars	0.125	0.0000
15	Kansas City Chiefs	0.125	0.0000
16	Las Vegas Raiders	0.250	0.1875
17	Los Angeles Chargers	0.438	0.6250
18	Los Angeles Rams	0.469	0.4375
19	Miami Dolphins	0.438	0.5000
20	Minnesota Vikings	0.625	0.5000
21	New England Patriots	0.750	1.0000

22	New Orleans Saints	0.438	0.5625
23	New York Giants	0.563	0.8125
24	New York Jets	0.375	0.3125
25	Philadelphia Eagles	0.250	0.0000
26	Pittsburgh Steelers	0.500	0.5000
27	San Francisco 49ers	0.719	0.8125
28	Seattle Seahawks	0.688	0.9375
29	Tampa Bay Buccaneers	0.438	0.1875
30	Tennessee Titans	0.375	0.1250
31	Washington Commanders	0.625	0.8125

```
Predictions for 2013...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.625	0.6250
1	Atlanta Falcons	0.250	0.0625
2	Baltimore Ravens	0.500	0.3750
3	Buffalo Bills	0.375	0.3750
4	Carolina Panthers	0.750	0.9375
5	Chicago Bears	0.500	0.3750
6	Cincinnati Bengals	0.688	0.9375
7	Cleveland Browns	0.250	0.1250
8	Dallas Cowboys	0.500	0.5000
9	Denver Broncos	0.813	1.0000
10	Detroit Lions	0.438	0.6250
11	Green Bay Packers	0.531	0.8125
12	Houston Texans	0.125	0.1250
13	Indianapolis Colts	0.688	0.6250
14	Jacksonville Jaguars	0.250	0.0000
15	Kansas City Chiefs	[illegible]	[illegible]

15	Kansas City Chiefs	0.688	0.8750
16	Las Vegas Raiders	0.250	0.1875
17	Los Angeles Chargers	0.563	0.5625
18	Los Angeles Rams	0.438	0.3750
19	Miami Dolphins	0.500	0.5000
20	Minnesota Vikings	0.344	0.0625
21	New England Patriots	0.750	0.8750
22	New Orleans Saints	0.688	0.6875
23	New York Giants	0.438	0.2500
24	New York Jets	0.500	0.1875
25	Philadelphia Eagles	0.625	0.8125
26	Pittsburgh Steelers	0.500	0.7500
27	San Francisco 49ers	0.750	0.8125
28	Seattle Seahawks	0.813	1.0000
29	Tampa Bay Buccaneers	0.250	0.1875
30	Tennessee Titans	0.438	0.3750
31	Washington Commanders	0.188	0.0000

```
Predictions for 2014...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.688	0.6250
1	Atlanta Falcons	0.375	0.5000
2	Baltimore Ravens	0.625	0.9375
3	Buffalo Bills	0.563	0.7500
4	Carolina Panthers	0.469	0.3125
5	Chicago Bears	0.313	0.1250
6	Cincinnati Bengals	0.656	0.5000
7	Cleveland Browns	0.438	0.2500
8	Dallas Cowboys	0.750	0.9375

9	Denver Broncos	0.750	0.9375
10	Detroit Lions	0.688	0.6875
11	Green Bay Packers	0.750	0.9375
12	Houston Texans	0.563	0.5625
13	Indianapolis Colts	0.688	0.8125
14	Jacksonville Jaguars	0.188	0.1250
15	Kansas City Chiefs	0.563	0.6875
16	Las Vegas Raiders	0.188	0.0000
17	Los Angeles Chargers	0.563	0.3750
18	Los Angeles Rams	0.375	0.2500
19	Miami Dolphins	0.500	0.4375
20	Minnesota Vikings	0.438	0.3125
21	New England Patriots	0.750	1.0000
22	New Orleans Saints	0.438	0.6250
23	New York Giants	0.375	0.3750
24	New York Jets	0.250	0.1250
25	Philadelphia Eagles	0.625	0.6875
26	Pittsburgh Steelers	0.688	0.7500
27	San Francisco 49ers	0.500	0.3125
28	Seattle Seahawks	0.750	0.8750
29	Tampa Bay Buccaneers	0.125	0.0625
30	Tennessee Titans	0.125	0.0000
31	Washington Commanders	0.250	0.1250

```
Predictions for 2015...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.813	1.0000
1	Atlanta Falcons	0.500	0.6875
2	Baltimore Ravens	0.313	0.3125

2	Baltimore Ravens	0.313	0.3125
3	Buffalo Bills	0.500	0.5000
4	Carolina Panthers	0.938	1.0000
5	Chicago Bears	0.375	0.1250
6	Cincinnati Bengals	0.750	0.8750
7	Cleveland Browns	0.188	0.0625
8	Dallas Cowboys	0.250	0.0000
9	Denver Broncos	0.750	0.6875
10	Detroit Lions	0.438	0.4375
11	Green Bay Packers	0.625	0.5625
12	Houston Texans	0.563	0.5625
13	Indianapolis Colts	0.500	0.1875
14	Jacksonville Jaguars	0.313	0.5625
15	Kansas City Chiefs	0.688	0.8750
16	Las Vegas Raiders	0.438	0.3750
17	Los Angeles Chargers	0.250	0.0625
18	Los Angeles Rams	0.438	0.4375
19	Miami Dolphins	0.375	0.1875
20	Minnesota Vikings	0.688	0.7500
21	New England Patriots	0.750	1.0000
22	New Orleans Saints	0.438	0.3750
23	New York Giants	0.375	0.6250
24	New York Jets	0.625	0.8125
25	Philadelphia Eagles	0.438	0.1875
26	Pittsburgh Steelers	0.625	0.6875
27	San Francisco 49ers	0.313	0.0000
28	Seattle Seahawks	0.625	0.8125
29	Tampa Bay Buccaneers	0.375	0.3125

30	Tennessee Titans	0.188	0.0625
31	Washington Commanders	0.563	0.8750

```
Predictions for 2016...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.469	0.8750
1	Atlanta Falcons	0.688	1.0000
2	Baltimore Ravens	0.500	0.4375
3	Buffalo Bills	0.438	0.6875
4	Carolina Panthers	0.375	0.1250
5	Chicago Bears	0.188	0.0625
6	Cincinnati Bengals	0.406	0.3125
7	Cleveland Browns	0.063	0.0000
8	Dallas Cowboys	0.813	1.0000
9	Denver Broncos	0.563	0.8750
10	Detroit Lions	0.563	0.3125
11	Green Bay Packers	0.625	0.8750
12	Houston Texans	0.563	0.1875
13	Indianapolis Colts	0.500	0.6875
14	Jacksonville Jaguars	0.188	0.0625
15	Kansas City Chiefs	0.750	0.8125
16	Las Vegas Raiders	0.750	0.6250
17	Los Angeles Chargers	0.313	0.2500
18	Los Angeles Rams	0.250	0.0000
19	Miami Dolphins	0.625	0.3750
20	Minnesota Vikings	0.500	0.4375
21	New England Patriots	0.875	1.0000
22	New Orleans Saints	0.438	0.5000
23	New York Giants	0.688	0.7500

24	New York Jets	0.313	0.1875
25	Philadelphia Eagles	0.438	0.5000
26	Pittsburgh Steelers	0.688	0.8125
27	San Francisco 49ers	0.125	0.1250
28	Seattle Seahawks	0.656	0.6875
29	Tampa Bay Buccaneers	0.563	0.3750
30	Tennessee Titans	0.563	0.6250
31	Washington Commanders	0.531	0.4375

```
Predictions for 2017...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.500	0.3125
1	Atlanta Falcons	0.625	0.7500
2	Baltimore Ravens	0.563	0.7500
3	Buffalo Bills	0.563	0.5000
4	Carolina Panthers	0.688	0.5625
5	Chicago Bears	0.313	0.3750
6	Cincinnati Bengals	0.438	0.3125
7	Cleveland Browns	0.000	0.0000
8	Dallas Cowboys	0.563	0.5625
9	Denver Broncos	0.313	0.1875
10	Detroit Lions	0.563	0.5625
11	Green Bay Packers	0.438	0.1250
12	Houston Texans	0.250	0.1875
13	Indianapolis Colts	0.250	0.0625
14	Jacksonville Jaguars	0.625	0.9375
15	Kansas City Chiefs	0.625	0.8125
16	Las Vegas Raiders	0.375	0.3125

17	Los Angeles Chargers	0.563	0.6875
18	Los Angeles Rams	0.688	0.9375
19	Miami Dolphins	0.375	0.0000
20	Minnesota Vikings	0.813	0.9375
21	New England Patriots	0.813	1.0000
22	New Orleans Saints	0.688	0.8125
23	New York Giants	0.188	0.0000
24	New York Jets	0.313	0.2500
25	Philadelphia Eagles	0.813	1.0000
26	Pittsburgh Steelers	0.813	0.8125
27	San Francisco 49ers	0.375	0.1875
28	Seattle Seahawks	0.563	0.6875
29	Tampa Bay Buccaneers	0.313	0.3750
30	Tennessee Titans	0.563	0.6250
31	Washington Commanders	0.438	0.3750

```
Predictions for 2018...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.188	0.0000
1	Atlanta Falcons	0.438	0.4375
2	Baltimore Ravens	0.625	0.8750
3	Buffalo Bills	0.375	0.0000
4	Carolina Panthers	0.438	0.5625
5	Chicago Bears	0.750	0.9375
6	Cincinnati Bengals	0.375	0.1250
7	Cleveland Browns	0.469	0.3125
8	Dallas Cowboys	0.625	0.6250
9	Denver Broncos	0.375	0.4375
10	Detroit Lions	0.375	0.3125

11	Green Bay Packers	0.406	0.5625
12	Houston Texans	0.688	0.8125
13	Indianapolis Colts	0.625	0.9375
14	Jacksonville Jaguars	0.313	0.2500
15	Kansas City Chiefs	0.750	1.0000
16	Las Vegas Raiders	0.250	0.0625
17	Los Angeles Chargers	0.750	0.7500
18	Los Angeles Rams	0.813	0.9375
19	Miami Dolphins	0.438	0.3125
20	Minnesota Vikings	0.531	0.5625
21	New England Patriots	0.688	0.8750
22	New Orleans Saints	0.813	0.9375
23	New York Giants	0.313	0.3125
24	New York Jets	0.250	0.1250
25	Philadelphia Eagles	0.563	0.7500
26	Pittsburgh Steelers	0.594	0.6250
27	San Francisco 49ers	0.250	0.1875
28	Seattle Seahawks	0.625	0.6875
29	Tampa Bay Buccaneers	0.313	0.1875
30	Tennessee Titans	0.563	0.4375
31	Washington Commanders	0.438	0.0625

```
Predictions for 2019...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.344	0.2500
1	Atlanta Falcons	0.438	0.2500
2	Baltimore Ravens	0.875	1.0000
3	Buffalo Bills	0.625	0.7500

4	Carolina Panthers	0.313	0.0625
5	Chicago Bears	0.500	0.4375
6	Cincinnati Bengals	0.125	0.0625
7	Cleveland Browns	0.375	0.3125
8	Dallas Cowboys	0.500	0.8750
9	Denver Broncos	0.438	0.3125
10	Detroit Lions	0.219	0.1875
11	Green Bay Packers	0.813	0.6875
12	Houston Texans	0.625	0.4375
13	Indianapolis Colts	0.438	0.6250
14	Jacksonville Jaguars	0.375	0.1875
15	Kansas City Chiefs	0.750	0.8750
16	Las Vegas Raiders	0.438	0.0625
17	Los Angeles Chargers	0.313	0.4375
18	Los Angeles Rams	0.563	0.6875
19	Miami Dolphins	0.313	0.0625
20	Minnesota Vikings	0.625	0.8750
21	New England Patriots	0.750	0.9375
22	New Orleans Saints	0.813	0.9375
23	New York Giants	0.250	0.1875
24	New York Jets	0.438	0.4375
25	Philadelphia Eagles	0.563	0.5625
26	Pittsburgh Steelers	0.500	0.5625
27	San Francisco 49ers	0.813	0.9375
28	Seattle Seahawks	0.688	0.5625
29	Tampa Bay Buccaneers	0.438	0.6250
30	Tennessee Titans	0.563	0.8125
31	Washington Commanders	0.188	0.0000

```
Predictions for 2020...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.500	0.6250
1	Atlanta Falcons	0.250	0.5000
2	Baltimore Ravens	0.688	1.0000
3	Buffalo Bills	0.813	1.0000
4	Carolina Panthers	0.313	0.1250
5	Chicago Bears	0.500	0.5625
6	Cincinnati Bengals	0.281	0.0625
7	Cleveland Browns	0.688	0.6250
8	Dallas Cowboys	0.375	0.3125
9	Denver Broncos	0.313	0.0625
10	Detroit Lions	0.313	0.0625
11	Green Bay Packers	0.813	0.9375
12	Houston Texans	0.250	0.2500
13	Indianapolis Colts	0.688	0.6875
14	Jacksonville Jaguars	0.063	0.0000
15	Kansas City Chiefs	0.875	0.7500
16	Las Vegas Raiders	0.500	0.3750
17	Los Angeles Chargers	0.438	0.3750
18	Los Angeles Rams	0.625	0.8750
19	Miami Dolphins	0.625	0.6875
20	Minnesota Vikings	0.438	0.3750
21	New England Patriots	0.438	0.3750
22	New Orleans Saints	0.750	1.0000
23	New York Giants	0.375	0.1875
24	New York Jets	0.125	0.0000

25	Philadelphia Eagles	0.281	0.0625
26	Pittsburgh Steelers	0.750	0.8125
27	San Francisco 49ers	0.375	0.3750
28	Seattle Seahawks	0.750	0.8125
29	Tampa Bay Buccaneers	0.688	0.8125
30	Tennessee Titans	0.688	0.7500
31	Washington Commanders	0.438	0.5625

```
Predictions for 2021...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.647	0.705882
1	Atlanta Falcons	0.412	0.117647
2	Baltimore Ravens	0.471	0.588235
3	Buffalo Bills	0.647	1.000000
4	Carolina Panthers	0.294	0.352941
5	Chicago Bears	0.353	0.176471
6	Cincinnati Bengals	0.588	0.941176
7	Cleveland Browns	0.471	0.352941
8	Dallas Cowboys	0.706	1.000000
9	Denver Broncos	0.412	0.529412
10	Detroit Lions	0.206	0.058824
11	Green Bay Packers	0.765	0.823529
12	Houston Texans	0.235	0.117647
13	Indianapolis Colts	0.529	0.823529
14	Jacksonville Jaguars	0.176	0.000000
15	Kansas City Chiefs	0.706	0.882353
16	Las Vegas Raiders	0.588	0.176471
17	Los Angeles Chargers	0.529	0.647059
18	Los Angeles Rams	0.706	0.882353

19	Miami Dolphins	0.529	0.470588
20	Minnesota Vikings	0.471	0.411765
21	New England Patriots	0.588	0.764706
22	New Orleans Saints	0.529	0.529412
23	New York Giants	0.235	0.000000
24	New York Jets	0.235	0.176471
25	Philadelphia Eagles	0.529	0.764706
26	Pittsburgh Steelers	0.559	0.176471
27	San Francisco 49ers	0.588	0.470588
28	Seattle Seahawks	0.412	0.411765
29	Tampa Bay Buccaneers	0.765	0.882353
30	Tennessee Titans	0.706	0.588235
31	Washington Commanders	0.412	0.176471

```
Predictions for 2022...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.235	0.000000
1	Atlanta Falcons	0.412	0.705882
2	Baltimore Ravens	0.588	0.823529
3	Buffalo Bills	0.813	1.000000
4	Carolina Panthers	0.412	0.529412
5	Chicago Bears	0.176	0.000000
6	Cincinnati Bengals	0.750	0.875000
7	Cleveland Browns	0.412	0.176471
8	Dallas Cowboys	0.706	0.882353
9	Denver Broncos	0.294	0.176471
10	Detroit Lions	0.529	0.764706
11	Green Bay Packers	0.471	0.470588

12	Houston Texans	0.206	0.176471
13	Indianapolis Colts	0.265	0.000000
14	Jacksonville Jaguars	0.529	0.647059
15	Kansas City Chiefs	0.824	0.882353
16	Las Vegas Raiders	0.353	0.529412
17	Los Angeles Chargers	0.588	0.705882
18	Los Angeles Rams	0.294	0.176471
19	Miami Dolphins	0.529	0.352941
20	Minnesota Vikings	0.765	0.352941
21	New England Patriots	0.471	0.588235
22	New Orleans Saints	0.412	0.235294
23	New York Giants	0.559	0.529412
24	New York Jets	0.412	0.764706
25	Philadelphia Eagles	0.824	1.000000
26	Pittsburgh Steelers	0.529	0.294118
27	San Francisco 49ers	0.765	1.000000
28	Seattle Seahawks	0.529	0.705882
29	Tampa Bay Buccaneers	0.471	0.117647
30	Tennessee Titans	0.412	0.294118
31	Washington Commanders	0.500	0.294118

```
Predictions for 2023...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.235	0.117647
1	Atlanta Falcons	0.412	0.352941
2	Baltimore Ravens	0.765	1.000000
3	Buffalo Bills	0.647	0.941176
4	Carolina Panthers	0.118	0.000000
5	Chicago Bears	0.412	0.529412

6	Cincinnati Bengals	0.529	0.411765
7	Cleveland Browns	0.647	0.823529
8	Dallas Cowboys	0.706	0.941176
9	Denver Broncos	0.471	0.176471
10	Detroit Lions	0.706	0.882353
11	Green Bay Packers	0.529	0.647059
12	Houston Texans	0.588	0.529412
13	Indianapolis Colts	0.529	0.352941
14	Jacksonville Jaguars	0.529	0.470588
15	Kansas City Chiefs	0.647	0.823529
16	Las Vegas Raiders	0.471	0.411765
17	Los Angeles Chargers	0.294	0.235294
18	Los Angeles Rams	0.588	0.647059
19	Miami Dolphins	0.647	0.764706
20	Minnesota Vikings	0.412	0.294118
21	New England Patriots	0.235	0.058824
22	New Orleans Saints	0.529	0.941176
23	New York Giants	0.353	0.176471
24	New York Jets	0.412	0.235294
25	Philadelphia Eagles	0.647	0.529412
26	Pittsburgh Steelers	0.588	0.647059
27	San Francisco 49ers	0.706	0.941176
28	Seattle Seahawks	0.529	0.352941
29	Tampa Bay Buccaneers	0.529	0.647059
30	Tennessee Titans	0.353	0.117647
31	Washington Commanders	0.235	0.000000

```
Predictions for 2024...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.471	0.647059
1	Atlanta Falcons	0.471	0.411765
2	Baltimore Ravens	0.706	0.941176
3	Buffalo Bills	0.765	0.941176
4	Carolina Panthers	0.294	0.000000
5	Chicago Bears	0.294	0.352941
6	Cincinnati Bengals	0.529	0.588235
7	Cleveland Browns	0.176	0.000000
8	Dallas Cowboys	0.412	0.235294
9	Denver Broncos	0.588	0.882353
10	Detroit Lions	0.882	1.000000
11	Green Bay Packers	0.647	0.823529
12	Houston Texans	0.588	0.647059
13	Indianapolis Colts	0.471	0.411765
14	Jacksonville Jaguars	0.235	0.294118
15	Kansas City Chiefs	0.882	0.588235
16	Las Vegas Raiders	0.235	0.117647
17	Los Angeles Chargers	0.647	0.764706
18	Los Angeles Rams	0.588	0.470588
19	Miami Dolphins	0.471	0.588235
20	Minnesota Vikings	0.824	0.764706
21	New England Patriots	0.235	0.058824
22	New Orleans Saints	0.294	0.352941
23	New York Giants	0.176	0.117647
24	New York Jets	0.294	0.294118
25	Philadelphia Eagles	0.824	0.941176
26	Pittsburgh Steelers	0.588	0.470588

27	San Francisco 49ers	0.353	0.117647
28	Seattle Seahawks	0.588	0.588235
29	Tampa Bay Buccaneers	0.588	0.823529
30	Tennessee Titans	0.176	0.000000
31	Washington Commanders	0.706	0.764706

```
Predictions for 2025...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.176	0.058824
1	Atlanta Falcons	0.471	0.588235
2	Baltimore Ravens	0.471	0.647059
3	Buffalo Bills	0.706	0.823529
4	Carolina Panthers	0.471	0.294118
5	Chicago Bears	0.647	0.764706
6	Cincinnati Bengals	0.353	0.235294
7	Cleveland Browns	0.294	0.176471
8	Dallas Cowboys	0.441	0.470588
9	Denver Broncos	0.824	0.823529
10	Detroit Lions	0.529	0.941176
11	Green Bay Packers	0.559	0.470588
12	Houston Texans	0.706	0.882353
13	Indianapolis Colts	0.471	0.647059
14	Jacksonville Jaguars	0.765	0.764706
15	Kansas City Chiefs	0.353	0.470588
16	Las Vegas Raiders	0.176	0.000000
17	Los Angeles Chargers	0.647	0.470588
18	Los Angeles Rams	0.706	0.882353
19	Miami Dolphins	0.412	0.352941
20	Minnesota Vikings	0.529	0.588235

20	Minnesota Vikings	[illegible]	[illegible]
21	New England Patriots	0.824	1.000000
22	New Orleans Saints	0.353	0.176471
23	New York Giants	0.235	0.235294
24	New York Jets	0.176	0.000000
25	Philadelphia Eagles	0.647	0.764706
26	Pittsburgh Steelers	0.588	0.352941
27	San Francisco 49ers	0.706	0.588235
28	Seattle Seahawks	0.824	1.000000
29	Tampa Bay Buccaneers	0.471	0.411765
30	Tennessee Titans	0.176	0.058824
31	Washington Commanders	0.294	0.058824

Next steps: Generate code with `preds` New interactive sheet Generate code with `preds`

```
from sklearn.metrics import r2_score, mean_squared_error

# Normalize test set with the same stats used during training
X_test_norm = (X_test_global - mean) / std

# Raw predictions from the neural network
y_pred_nn = model.predict(X_test_norm, verbose=0).flatten()

r2  = r2_score(y_test_global, y_pred_nn)
mse = mean_squared_error(y_test_global, y_pred_nn)

# Percent error per sample (avoid divide-by-zero for 0.0 actual values
with np.errstate(divide='ignore', invalid='ignore'):
    pct_err = np.where(
        y_test_global != 0,
        np.abs((y_test_global - y_pred_nn) / y_test_global) * 100,
        np.nan
    )
mean_pct_err = np.nanmean(pct_err)

print(f"R²: {r2:.4f}")
print(f"MSE: {mse:.6f}")
print(f"Mean % Error: {mean_pct_err:.2f} %")
```

```
R²: 0.8085
MSE: 0.007477
Mean % Error: 17.37 %
```

Random Forest Model

Simon Liedtke

```
import os
import pickle
import numpy as np
import pandas as pd
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import RandomizedSearchCV
from sklearn.metrics import mean_squared_error, r2_score, mean_absolu

RF_MODEL_FILEPATH = "Models/Random Forest"
RF_PREDICTIONS_FILEPATH = "Predictions/Random Forest"

SCHEDULE_FILEPATH = "Schedule"
DATA_FILEPATH = "Data"

TEAM = 0; WINS = 1; LOSSES = 2; TIES = 3; WIN_PERCENTAGE = 4
POINTS_FOR = 5; POINTS_AGAINST = 6; POINTS_DIFFERENTIAL = 7
MARGIN_OF_VICTORY = 8; STRENGTH_OF_SCHEDULE = 9
SIMPLE_RATING_SYSTEM = 10; OFFENSIVE_SRS = 11; DEFENSIVE_SRS = 12

NFL_TEAMS = ['Arizona Cardinals', 'Atlanta Falcons', 'Baltimore Raven
             'Carolina Panthers', 'Chicago Bears', 'Cincinnati Bengal
             'Dallas Cowboys', 'Denver Broncos', 'Detroit Lions', 'Gr
             'Houston Texans', 'Indianapolis Colts', 'Jacksonville Ja
             'Las Vegas Raiders', 'Los Angeles Chargers', 'Los Angele
             'Minnesota Vikings', 'New England Patriots', 'New Orlean
             'New York Jets', 'Philadelphia Eagles', 'Pittsburgh Stee
             'Seattle Seahawks', 'Tampa Bay Buccaneers', 'Tennessee T
NFL_YEARS = [str(year) for year in range(2010, 2026)]
DROP_COLS = ['W', 'L', 'T', 'W-L%', 'PF', 'PA', 'PD', 'MoV']
```

```
def read_data(NFL_YEAR, team_name):
    team_data = []
    with open(os.path.join(DATA_FILEPATH, f"{NFL_YEAR}.csv"), "r") as
        data = pd.read_csv(file)
        for index, row in data.iterrows():
            if row.iloc[TEAM] == team_name:
                team_data.append([
                    row.iloc[WINS], row.iloc[LOSSES], row.iloc[TIES],
                    row.iloc[POINTS_FOR], row.iloc[POINTS_AGAINST], r
                    row.iloc[MARGIN_OF_VICTORY], row.iloc[STRENGTH_OF_
                    row.iloc[SIMPLE_RATING_SYSTEM], row.iloc[OFFENSIVI
                ])
    return pd.DataFrame(team_data, columns=['W', 'L', 'T', 'W-L%', 'PI
```

```
def train_model():
    param_grid = {
        'n_estimators':      [5000],
        'max_depth':         [None, 2, 20],
        'min_samples_split': [2, 20],
        'min_samples_leaf':  [2, 20],
        'bootstrap':         [True, False],
    }
    random_search = RandomizedSearchCV(
        RandomForestRegressor(random_state=42),
        param_grid,
        n_iter=10,
        cv=3,
        random_state=42,
        n_jobs=-1
    )
    random_search.fit(X_train_global, y_train_global)
    print("Best hyperparameters:", random_search.best_params_)
    return random_search.best_estimator_
```

```
def save_model(NFL_YEAR, model):
    os.makedirs(RF_MODEL_FILEPATH, exist_ok=True)
    with open(os.path.join(RF_MODEL_FILEPATH, f"rf_model_{NFL_YEAR}.pl
        pickle.dump(model, f)
```

```
def make_predictions(NFL_YEAR):
```

```
def make_predictions(NFL_YEAR):
    model_path = os.path.join(RF_MODEL_FILEPATH, "rf_model_global.pkl'
    with open(model_path, "rb") as f:
        model = pickle.load(f)

    schedule_df = pd.read_csv(os.path.join(SCHEDULE_FILEPATH, f"{NFL_'
    if schedule_df.empty:
        print(f"No schedule found for {NFL_YEAR}")
        return

    try:
        season_data = pd.read_csv(os.path.join(DATA_FILEPATH, f"{NFL_'
        team_stats  = {row.iloc[TEAM]: read_data(NFL_YEAR, row.iloc[TI
                       for _, row in season_data.iterrows()}
        team_stats  = {k: v for k, v in team_stats.items() if v is no
    except FileNotFoundError:
        print(f"Data file for {NFL_YEAR} not found")
        return

    team_wins  = {team: 0 for team in NFL_TEAMS}
    team_games = {team: 0 for team in NFL_TEAMS}

    for _, row in schedule_df.iterrows():
        winner, loser = row['Winner/tie'], row['Loser/tie']
        if winner not in team_stats or loser not in team_stats:
            continue
        winner_pred = model.predict(team_stats[winner].drop(columns=DI
        loser_pred  = model.predict(team_stats[loser].drop(columns=DR(
        predicted_winner = winner if winner_pred > loser_pred else lo
        predicted_loser  = loser  if winner_pred > loser_pred else wi
        team_wins[predicted_winner]  += 1
        team_games[predicted_winner] += 1
        team_games[predicted_loser]  += 1

    predictions = [
        [team,
         team_stats[team]['W-L%'].iloc[0] if team in team_stats else (
         team_wins[team] / team_games[team] if team_games[team] > 0 e
        for team in NFL_TEAMS
    ]
    predictions_df = pd.DataFrame(predictions, columns=['Team', 'Actu
    os.makedirs(RF_PREDICTIONS_FILEPATH, exist_ok=True)
    predictions_df.to_csv(os.path.join(RF_PREDICTIONS_FILEPATH, f"pre
    return predictions_df
```

```
rf_model = train_model()
save_model("global", rf_model)
print("Model saved.")

# Evaluate on the global validation split
y_pred_rf_val = rf_model.predict(X_val_global)
print(f"R²:  {r2_score(y_val_global,  y_pred_rf_val):.4f}")
print(f"MSE: {mean_squared_error(y_val_global, y_pred_rf_val):.6f}")

# Run game-by-game predictions for each year using the global model
for year in NFL_YEARS:
    print(f"\nPredictions for {year}...")
    preds = make_predictions(year)
    if preds is not None:
        display(preds)
```

```
Best hyperparameters: {'n_estimators': 5000, 'min_samples_split': 20,
Model saved.
R²:  0.8190
MSE: 0.006972

Predictions for 2010...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.313	0.0625
1	Atlanta Falcons	0.813	0.8750
2	Baltimore Ravens	0.750	0.7500
3	Buffalo Bills	0.250	0.0000
4	Carolina Panthers	0.125	0.0000
5	Chicago Bears	0.688	0.7500
6	Cincinnati Bengals	0.250	0.1250
7	Cleveland Browns	0.313	0.3125
8	Dallas Cowboys	0.375	0.3125
9	Denver Broncos	0.250	0.0625
10	Detroit Lions	0.375	0.4375
11	Green Bay Packers	0.625	0.9375

12	Houston Texans	0.375	0.3125
13	Indianapolis Colts	0.625	0.8125
14	Jacksonville Jaguars	0.500	0.1250
15	Kansas City Chiefs	0.625	0.6875
16	Las Vegas Raiders	0.500	0.7500
17	Los Angeles Chargers	0.563	0.9375
18	Los Angeles Rams	0.438	0.3750
19	Miami Dolphins	0.438	0.3125
20	Minnesota Vikings	0.375	0.1250
21	New England Patriots	0.875	1.0000
22	New Orleans Saints	0.688	0.7500
23	New York Giants	0.625	0.6875
24	New York Jets	0.688	0.6875
25	Philadelphia Eagles	0.625	0.8125
26	Pittsburgh Steelers	0.750	0.9375
27	San Francisco 49ers	0.375	0.5000
28	Seattle Seahawks	0.438	0.2500
29	Tampa Bay Buccaneers	0.625	0.6250
30	Tennessee Titans	0.375	0.5000
31	Washington Commanders	0.375	0.1875

```
Predictions for 2011...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.500	0.3750
1	Atlanta Falcons	0.625	0.6875
2	Baltimore Ravens	0.750	1.0000
3	Buffalo Bills	0.375	0.2500
4	Carolina Panthers	0.375	0.3750
5	Chicago Bears	0.500	0.5000

6	Cincinnati Bengals	0.563	0.6250
7	Cleveland Browns	0.250	0.1875
8	Dallas Cowboys	0.500	0.6875
9	Denver Broncos	0.500	0.3125
10	Detroit Lions	0.625	0.7500
11	Green Bay Packers	0.938	1.0000
12	Houston Texans	0.625	0.8125
13	Indianapolis Colts	0.125	0.0000
14	Jacksonville Jaguars	0.313	0.2500
15	Kansas City Chiefs	0.438	0.0625
16	Las Vegas Raiders	0.500	0.1875
17	Los Angeles Chargers	0.500	0.7500
18	Los Angeles Rams	0.125	0.0000
19	Miami Dolphins	0.375	0.4375
20	Minnesota Vikings	0.188	0.1250
21	New England Patriots	0.813	1.0000
22	New Orleans Saints	0.813	0.9375
23	New York Giants	0.563	0.5000
24	New York Jets	0.500	0.5625
25	Philadelphia Eagles	0.500	0.8125
26	Pittsburgh Steelers	0.750	0.7500
27	San Francisco 49ers	0.813	0.9375
28	Seattle Seahawks	0.438	0.3750
29	Tampa Bay Buccaneers	0.250	0.0625
30	Tennessee Titans	0.563	0.5625
31	Washington Commanders	0.313	0.1250

```
Predictions for 2012...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.313	0.1250
1	Atlanta Falcons	0.813	0.9375
2	Baltimore Ravens	0.625	0.6875
3	Buffalo Bills	0.375	0.1875
4	Carolina Panthers	0.438	0.2500
5	Chicago Bears	0.625	0.8750
6	Cincinnati Bengals	0.625	0.6875
7	Cleveland Browns	0.313	0.2500
8	Dallas Cowboys	0.500	0.1875
9	Denver Broncos	0.813	0.9375
10	Detroit Lions	0.250	0.2500
11	Green Bay Packers	0.688	0.7500
12	Houston Texans	0.750	0.7500
13	Indianapolis Colts	0.688	0.5625
14	Jacksonville Jaguars	0.125	0.0000
15	Kansas City Chiefs	0.125	0.0000
16	Las Vegas Raiders	0.250	0.1875
17	Los Angeles Chargers	0.438	0.6250
18	Los Angeles Rams	0.469	0.3125
19	Miami Dolphins	0.438	0.6250
20	Minnesota Vikings	0.625	0.5000
21	New England Patriots	0.750	1.0000
22	New Orleans Saints	0.438	0.5000
23	New York Giants	0.563	0.8125
24	New York Jets	0.375	0.3125
25	Philadelphia Eagles	0.250	0.0000
26	Pittsburgh Steelers	0.500	0.5625

27	San Francisco 49ers	0.719	0.8125
28	Seattle Seahawks	0.688	0.9375
29	Tampa Bay Buccaneers	0.438	0.4375
30	Tennessee Titans	0.375	0.1250
31	Washington Commanders	0.625	0.8125

```
Predictions for 2013...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.625	0.6250
1	Atlanta Falcons	0.250	0.0625
2	Baltimore Ravens	0.500	0.5000
3	Buffalo Bills	0.375	0.3750
4	Carolina Panthers	0.750	0.8750
5	Chicago Bears	0.500	0.3125
6	Cincinnati Bengals	0.688	0.9375
7	Cleveland Browns	0.250	0.0625
8	Dallas Cowboys	0.500	0.6250
9	Denver Broncos	0.813	1.0000
10	Detroit Lions	0.438	0.6875
11	Green Bay Packers	0.531	0.5625
12	Houston Texans	0.125	0.1250
13	Indianapolis Colts	0.688	0.6250
14	Jacksonville Jaguars	0.250	0.0000
15	Kansas City Chiefs	0.688	0.8750
16	Las Vegas Raiders	0.250	0.1875
17	Los Angeles Chargers	0.563	0.6250
18	Los Angeles Rams	0.438	0.3750
19	Miami Dolphins	0.500	0.4375
20	Minnesota Vikings	0.344	0.1875

20	Minnesota Vikings	0.344	0.1875
21	New England Patriots	0.750	0.8750
22	New Orleans Saints	0.688	0.6875
23	New York Giants	0.438	0.1875
24	New York Jets	0.500	0.2500
25	Philadelphia Eagles	0.625	0.7500
26	Pittsburgh Steelers	0.500	0.8125
27	San Francisco 49ers	0.750	0.8750
28	Seattle Seahawks	0.813	1.0000
29	Tampa Bay Buccaneers	0.250	0.1250
30	Tennessee Titans	0.438	0.3750
31	Washington Commanders	0.188	0.0000

```
Predictions for 2014...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.688	0.5625
1	Atlanta Falcons	0.375	0.5000
2	Baltimore Ravens	0.625	1.0000
3	Buffalo Bills	0.563	0.6875
4	Carolina Panthers	0.469	0.3125
5	Chicago Bears	0.313	0.0625
6	Cincinnati Bengals	0.656	0.5000
7	Cleveland Browns	0.438	0.2500
8	Dallas Cowboys	0.750	0.9375
9	Denver Broncos	0.750	1.0000
10	Detroit Lions	0.688	0.7500
11	Green Bay Packers	0.750	0.8750
12	Houston Texans	0.563	0.5625
13	Indianapolis Colts	0.688	0.7500

14	Jacksonville Jaguars	0.188	0.1250
15	Kansas City Chiefs	0.563	0.7500
16	Las Vegas Raiders	0.188	0.0000
17	Los Angeles Chargers	0.563	0.3750
18	Los Angeles Rams	0.375	0.1875
19	Miami Dolphins	0.500	0.4375
20	Minnesota Vikings	0.438	0.3750
21	New England Patriots	0.750	0.9375
22	New Orleans Saints	0.438	0.6250
23	New York Giants	0.375	0.3125
24	New York Jets	0.250	0.1875
25	Philadelphia Eagles	0.625	0.6875
26	Pittsburgh Steelers	0.688	0.7500
27	San Francisco 49ers	0.500	0.3750
28	Seattle Seahawks	0.750	0.9375
29	Tampa Bay Buccaneers	0.125	0.0000
30	Tennessee Titans	0.125	0.0000
31	Washington Commanders	0.250	0.1875

```
Predictions for 2015...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.813	1.0000
1	Atlanta Falcons	0.500	0.6875
2	Baltimore Ravens	0.313	0.2500
3	Buffalo Bills	0.500	0.6250
4	Carolina Panthers	0.938	1.0000
5	Chicago Bears	0.375	0.1875
6	Cincinnati Bengals	0.750	0.8750
7	Cleveland Browns	0.188	0.0625

7	Cleveland Browns	0.188	0.0625
8	Dallas Cowboys	0.250	0.0000
9	Denver Broncos	0.750	0.6250
10	Detroit Lions	0.438	0.4375
11	Green Bay Packers	0.625	0.5625
12	Houston Texans	0.563	0.6250
13	Indianapolis Colts	0.500	0.4375
14	Jacksonville Jaguars	0.313	0.3125
15	Kansas City Chiefs	0.688	0.9375
16	Las Vegas Raiders	0.438	0.3750
17	Los Angeles Chargers	0.250	0.0625
18	Los Angeles Rams	0.438	0.4375
19	Miami Dolphins	0.375	0.3750
20	Minnesota Vikings	0.688	0.8125
21	New England Patriots	0.750	1.0000
22	New Orleans Saints	0.438	0.3750
23	New York Giants	0.375	0.5000
24	New York Jets	0.625	0.8750
25	Philadelphia Eagles	0.438	0.1875
26	Pittsburgh Steelers	0.625	0.6250
27	San Francisco 49ers	0.313	0.0625
28	Seattle Seahawks	0.625	0.8125
29	Tampa Bay Buccaneers	0.375	0.1250
30	Tennessee Titans	0.188	0.0000
31	Washington Commanders	0.563	0.7500

```
Predictions for 2016...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.469	0.7500

1	Atlanta Falcons	0.688	1.0000
2	Baltimore Ravens	0.500	0.3750
3	Buffalo Bills	0.438	0.6250
4	Carolina Panthers	0.375	0.1250
5	Chicago Bears	0.188	0.0625
6	Cincinnati Bengals	0.406	0.2500
7	Cleveland Browns	0.063	0.0000
8	Dallas Cowboys	0.813	1.0000
9	Denver Broncos	0.563	0.7500
10	Detroit Lions	0.563	0.3125
11	Green Bay Packers	0.625	0.8125
12	Houston Texans	0.563	0.1875
13	Indianapolis Colts	0.500	0.6875
14	Jacksonville Jaguars	0.188	0.0625
15	Kansas City Chiefs	0.750	0.9375
16	Las Vegas Raiders	0.750	0.6875
17	Los Angeles Chargers	0.313	0.2500
18	Los Angeles Rams	0.250	0.0000
19	Miami Dolphins	0.625	0.5000
20	Minnesota Vikings	0.500	0.4375
21	New England Patriots	0.875	1.0000
22	New Orleans Saints	0.438	0.5000
23	New York Giants	0.688	0.6250
24	New York Jets	0.313	0.1875
25	Philadelphia Eagles	0.438	0.7500
26	Pittsburgh Steelers	0.688	0.8125
27	San Francisco 49ers	0.125	0.1250
28	Seattle Seahawks	0.656	0.7500

29	Tampa Bay Buccaneers	0.563	0.3750
30	Tennessee Titans	0.563	0.6250
31	Washington Commanders	0.531	0.4375

```
Predictions for 2017...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.500	0.4375
1	Atlanta Falcons	0.625	0.7500
2	Baltimore Ravens	0.563	0.7500
3	Buffalo Bills	0.563	0.4375
4	Carolina Panthers	0.688	0.5625
5	Chicago Bears	0.313	0.1875
6	Cincinnati Bengals	0.438	0.5000
7	Cleveland Browns	0.000	0.0000
8	Dallas Cowboys	0.563	0.6250
9	Denver Broncos	0.313	0.1250
10	Detroit Lions	0.563	0.5625
11	Green Bay Packers	0.438	0.0625
12	Houston Texans	0.250	0.1875
13	Indianapolis Colts	0.250	0.0625
14	Jacksonville Jaguars	0.625	0.9375
15	Kansas City Chiefs	0.625	0.6875
16	Las Vegas Raiders	0.375	0.3125
17	Los Angeles Chargers	0.563	0.8125
18	Los Angeles Rams	0.688	0.9375
19	Miami Dolphins	0.375	0.0625
20	Minnesota Vikings	0.813	0.8750
21	New England Patriots	0.813	1.0000

22	New Orleans Saints	0.688	0.8750
23	New York Giants	0.188	0.0000
24	New York Jets	0.313	0.2500
25	Philadelphia Eagles	0.813	1.0000
26	Pittsburgh Steelers	0.813	0.8125
27	San Francisco 49ers	0.375	0.3125
28	Seattle Seahawks	0.563	0.6250
29	Tampa Bay Buccaneers	0.313	0.3750
30	Tennessee Titans	0.563	0.6250
31	Washington Commanders	0.438	0.2500

```
Predictions for 2018...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.188	0.0000
1	Atlanta Falcons	0.438	0.6250
2	Baltimore Ravens	0.625	0.8125
3	Buffalo Bills	0.375	0.1250
4	Carolina Panthers	0.438	0.4375
5	Chicago Bears	0.750	0.9375
6	Cincinnati Bengals	0.375	0.1250
7	Cleveland Browns	0.469	0.3125
8	Dallas Cowboys	0.625	0.5625
9	Denver Broncos	0.375	0.4375
10	Detroit Lions	0.375	0.3125
11	Green Bay Packers	0.406	0.5625
12	Houston Texans	0.688	0.9375
13	Indianapolis Colts	0.625	0.8125
14	Jacksonville Jaguars	0.313	0.2500
15	Kansas City Chiefs	0.750	1.0000

16	Las Vegas Raiders	0.250	0.0625
17	Los Angeles Chargers	0.750	0.8125
18	Los Angeles Rams	0.813	0.8750
19	Miami Dolphins	0.438	0.3125
20	Minnesota Vikings	0.531	0.6250
21	New England Patriots	0.688	0.8750
22	New Orleans Saints	0.813	1.0000
23	New York Giants	0.313	0.3125
24	New York Jets	0.250	0.0000
25	Philadelphia Eagles	0.563	0.6875
26	Pittsburgh Steelers	0.594	0.6250
27	San Francisco 49ers	0.250	0.1875
28	Seattle Seahawks	0.625	0.6875
29	Tampa Bay Buccaneers	0.313	0.1875
30	Tennessee Titans	0.563	0.4375
31	Washington Commanders	0.438	0.0625

```
Predictions for 2019...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.344	0.2500
1	Atlanta Falcons	0.438	0.3125
2	Baltimore Ravens	0.875	1.0000
3	Buffalo Bills	0.625	0.6875
4	Carolina Panthers	0.313	0.0625
5	Chicago Bears	0.500	0.4375
6	Cincinnati Bengals	0.125	0.0000
7	Cleveland Browns	0.375	0.2500
8	Dallas Cowboys	0.500	0.7500

9	Denver Broncos	0.438	0.3125
10	Detroit Lions	0.219	0.1875
11	Green Bay Packers	0.813	0.7500
12	Houston Texans	0.625	0.3750
13	Indianapolis Colts	0.438	0.7500
14	Jacksonville Jaguars	0.375	0.1875
15	Kansas City Chiefs	0.750	0.8750
16	Las Vegas Raiders	0.438	0.0625
17	Los Angeles Chargers	0.313	0.5000
18	Los Angeles Rams	0.563	0.6875
19	Miami Dolphins	0.313	0.1250
20	Minnesota Vikings	0.625	0.9375
21	New England Patriots	0.750	0.9375
22	New Orleans Saints	0.813	0.9375
23	New York Giants	0.250	0.1875
24	New York Jets	0.438	0.5000
25	Philadelphia Eagles	0.563	0.5625
26	Pittsburgh Steelers	0.500	0.4375
27	San Francisco 49ers	0.813	0.9375
28	Seattle Seahawks	0.688	0.5625
29	Tampa Bay Buccaneers	0.438	0.5625
30	Tennessee Titans	0.563	0.8750
31	Washington Commanders	0.188	0.0000

```
Predictions for 2020...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.500	0.6250
1	Atlanta Falcons	0.250	0.3125
2	Baltimore Ravens	0.688	1.0000

3	Buffalo Bills	0.813	1.0000
4	Carolina Panthers	0.313	0.1250
5	Chicago Bears	0.500	0.5625
6	Cincinnati Bengals	0.281	0.0625
7	Cleveland Browns	0.688	0.6250
8	Dallas Cowboys	0.375	0.1875
9	Denver Broncos	0.313	0.0625
10	Detroit Lions	0.313	0.0625
11	Green Bay Packers	0.813	0.8750
12	Houston Texans	0.250	0.2500
13	Indianapolis Colts	0.688	0.8125
14	Jacksonville Jaguars	0.063	0.0000
15	Kansas City Chiefs	0.875	0.7500
16	Las Vegas Raiders	0.500	0.4375
17	Los Angeles Chargers	0.438	0.4375
18	Los Angeles Rams	0.625	0.7500
19	Miami Dolphins	0.625	0.6875
20	Minnesota Vikings	0.438	0.4375
21	New England Patriots	0.438	0.4375
22	New Orleans Saints	0.750	1.0000
23	New York Giants	0.375	0.3125
24	New York Jets	0.125	0.0000
25	Philadelphia Eagles	0.281	0.0625
26	Pittsburgh Steelers	0.750	0.8125
27	San Francisco 49ers	0.375	0.2500
28	Seattle Seahawks	0.750	0.9375
29	Tampa Bay Buccaneers	0.688	0.8750

30	Tennessee Titans	0.688	0.6250
31	Washington Commanders	0.438	0.6250

```
Predictions for 2021...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.647	0.588235
1	Atlanta Falcons	0.412	0.235294
2	Baltimore Ravens	0.471	0.529412
3	Buffalo Bills	0.647	1.000000
4	Carolina Panthers	0.294	0.352941
5	Chicago Bears	0.353	0.176471
6	Cincinnati Bengals	0.588	0.823529
7	Cleveland Browns	0.471	0.235294
8	Dallas Cowboys	0.706	1.000000
9	Denver Broncos	0.412	0.588235
10	Detroit Lions	0.206	0.000000
11	Green Bay Packers	0.765	0.882353
12	Houston Texans	0.235	0.117647
13	Indianapolis Colts	0.529	0.764706
14	Jacksonville Jaguars	0.176	0.000000
15	Kansas City Chiefs	0.706	0.882353
16	Las Vegas Raiders	0.588	0.176471
17	Los Angeles Chargers	0.529	0.647059
18	Los Angeles Rams	0.706	0.705882
19	Miami Dolphins	0.529	0.470588
20	Minnesota Vikings	0.471	0.411765
21	New England Patriots	0.588	0.764706
22	New Orleans Saints	0.529	0.470588
23	New York Giants	0.235	0.000000

24	New York Jets	0.235	0.117647
25	Philadelphia Eagles	0.529	0.705882
26	Pittsburgh Steelers	0.559	0.294118
27	San Francisco 49ers	0.588	1.000000
28	Seattle Seahawks	0.412	0.470588
29	Tampa Bay Buccaneers	0.765	0.882353
30	Tennessee Titans	0.706	0.529412
31	Washington Commanders	0.412	0.176471

```
Predictions for 2022...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.235	0.000000
1	Atlanta Falcons	0.412	0.705882
2	Baltimore Ravens	0.588	0.823529
3	Buffalo Bills	0.813	1.000000
4	Carolina Panthers	0.412	0.470588
5	Chicago Bears	0.176	0.058824
6	Cincinnati Bengals	0.750	0.875000
7	Cleveland Browns	0.412	0.470588
8	Dallas Cowboys	0.706	0.882353
9	Denver Broncos	0.294	0.176471
10	Detroit Lions	0.529	0.764706
11	Green Bay Packers	0.471	0.411765
12	Houston Texans	0.206	0.117647
13	Indianapolis Colts	0.265	0.000000
14	Jacksonville Jaguars	0.529	0.764706
15	Kansas City Chiefs	0.824	0.882353
16	Las Vegas Raiders	0.353	0.529412

17	Los Angeles Chargers	0.588	0.764706
18	Los Angeles Rams	0.294	0.176471
19	Miami Dolphins	0.529	0.411765
20	Minnesota Vikings	0.765	0.705882
21	New England Patriots	0.471	0.647059
22	New Orleans Saints	0.412	0.294118
23	New York Giants	0.559	0.470588
24	New York Jets	0.412	0.117647
25	Philadelphia Eagles	0.824	1.000000
26	Pittsburgh Steelers	0.529	0.176471
27	San Francisco 49ers	0.765	1.000000
28	Seattle Seahawks	0.529	0.705882
29	Tampa Bay Buccaneers	0.471	0.117647
30	Tennessee Titans	0.412	0.294118
31	Washington Commanders	0.500	0.235294

```
Predictions for 2023...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.235	0.117647
1	Atlanta Falcons	0.412	0.294118
2	Baltimore Ravens	0.765	0.941176
3	Buffalo Bills	0.647	0.941176
4	Carolina Panthers	0.118	0.000000
5	Chicago Bears	0.412	0.470588
6	Cincinnati Bengals	0.529	0.411765
7	Cleveland Browns	0.647	0.823529
8	Dallas Cowboys	0.706	0.941176
9	Denver Broncos	0.471	0.294118
10	Detroit Lions	0.706	0.823529

11	Green Bay Packers	0.529	0.705882
12	Houston Texans	0.588	0.823529
13	Indianapolis Colts	0.529	0.352941
14	Jacksonville Jaguars	0.529	0.529412
15	Kansas City Chiefs	0.647	0.941176
16	Las Vegas Raiders	0.471	0.647059
17	Los Angeles Chargers	0.294	0.176471
18	Los Angeles Rams	0.588	0.705882
19	Miami Dolphins	0.647	0.705882
20	Minnesota Vikings	0.412	0.235294
21	New England Patriots	0.235	0.117647
22	New Orleans Saints	0.529	0.882353
23	New York Giants	0.353	0.000000
24	New York Jets	0.412	0.235294
25	Philadelphia Eagles	0.647	0.588235
26	Pittsburgh Steelers	0.588	0.117647
27	San Francisco 49ers	0.706	1.000000
28	Seattle Seahawks	0.529	0.411765
29	Tampa Bay Buccaneers	0.529	0.470588
30	Tennessee Titans	0.353	0.176471
31	Washington Commanders	0.235	0.117647

```
Predictions for 2024...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.471	0.647059
1	Atlanta Falcons	0.471	0.411765
2	Baltimore Ravens	0.706	1.000000
3	Buffalo Bills	0.765	0.882353
4	Carolina Panthers	0.294	0.000000

4	Carolina Panthers	0.294	0.000000
5	Chicago Bears	0.294	0.294118
6	Cincinnati Bengals	0.529	0.588235
7	Cleveland Browns	0.176	0.000000
8	Dallas Cowboys	0.412	0.235294
9	Denver Broncos	0.588	0.941176
10	Detroit Lions	0.882	1.000000
11	Green Bay Packers	0.647	0.705882
12	Houston Texans	0.588	0.647059
13	Indianapolis Colts	0.471	0.470588
14	Jacksonville Jaguars	0.235	0.235294
15	Kansas City Chiefs	0.882	0.764706
16	Las Vegas Raiders	0.235	0.176471
17	Los Angeles Chargers	0.647	0.647059
18	Los Angeles Rams	0.588	0.411765
19	Miami Dolphins	0.471	0.647059
20	Minnesota Vikings	0.824	0.882353
21	New England Patriots	0.235	0.058824
22	New Orleans Saints	0.294	0.352941
23	New York Giants	0.176	0.117647
24	New York Jets	0.294	0.235294
25	Philadelphia Eagles	0.824	0.941176
26	Pittsburgh Steelers	0.588	0.470588
27	San Francisco 49ers	0.353	0.176471
28	Seattle Seahawks	0.588	0.588235
29	Tampa Bay Buccaneers	0.588	0.705882
30	Tennessee Titans	0.176	0.000000
31	Washington Commanders	0.706	0.764706

```
Predictions for 2025...
```

	Team	Actual W-L%	Predicted W-L%
0	Arizona Cardinals	0.176	0.058824
1	Atlanta Falcons	0.471	0.470588
2	Baltimore Ravens	0.471	0.588235
3	Buffalo Bills	0.706	0.823529
4	Carolina Panthers	0.471	0.352941
5	Chicago Bears	0.647	0.764706
6	Cincinnati Bengals	0.353	0.235294
7	Cleveland Browns	0.294	0.176471
8	Dallas Cowboys	0.441	0.294118
9	Denver Broncos	0.824	0.941176
10	Detroit Lions	0.529	0.941176
11	Green Bay Packers	0.559	0.647059
12	Houston Texans	0.706	0.705882
13	Indianapolis Colts	0.471	0.529412
14	Jacksonville Jaguars	0.765	0.882353
15	Kansas City Chiefs	0.353	0.352941
16	Las Vegas Raiders	0.176	0.000000
17	Los Angeles Chargers	0.647	0.647059
18	Los Angeles Rams	0.706	0.882353
19	Miami Dolphins	0.412	0.352941
20	Minnesota Vikings	0.529	0.352941
21	New England Patriots	0.824	1.000000
22	New Orleans Saints	0.353	0.176471
23	New York Giants	0.235	0.352941
24	New York Jets	0.176	0.000000
25	Philadelphia Eagles	0.647	0.764706

26	Pittsburgh Steelers	0.588	0.411765
27	San Francisco 49ers	0.706	0.647059
28	Seattle Seahawks	0.824	1.000000
29	Tampa Bay Buccaneers	0.471	0.529412
30	Tennessee Titans	0.176	0.058824
31	Washington Commanders	0.294	0.058824

Next steps: Generate code with preds | New interactive sheet | Generate code with preds

```
# Evaluate on the global test split (held out until final evaluation)
y_pred_rf_test = rf_model.predict(X_test_global)

mse = mean_squared_error(y_test_global, y_pred_rf_test)
r2  = r2_score(y_test_global, y_pred_rf_test)
mae = mean_absolute_error(y_test_global, y_pred_rf_test)

# Percent error per sample
with np.errstate(divide='ignore', invalid='ignore'):
    pct_err = np.where(
        y_test_global != 0,
        np.abs((y_test_global - y_pred_rf_test) / y_test_global) * 100,
        np.nan
    )
mean_pct_err = np.nanmean(pct_err)

print("Random Forest — Test-Set Performance (global 20% hold-out)")
print(f"  R²            : {r2:.4f}")
print(f"  MSE           : {mse:.6f}")
print(f"  MAE           : {mae:.6f}")
print(f"  Mean % Error  : {mean_pct_err:.2f} %")
```

```
Random Forest — Test-Set Performance (global 20% hold-out)
  R²            : 0.8078
  MSE           : 0.007503
  MAE           : 0.072904
  Mean % Error  : 19.18 %
```

Naive Bayes Model

Dilen Patel

```
import os
import joblib
import numpy as np
import pandas as pd
from sklearn.naive_bayes import GaussianNB
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import (
    accuracy_score,
    classification_report,
    confusion_matrix,
    mean_squared_error,
    r2_score,
)

NB_MODEL_FILEPATH       = "Models/Naive Bayes"
NB_PREDICTIONS_FILEPATH = "Predictions/Naive Bayes"
```

```
def wl_bucket_label(wlp):
    """Convert W-L% into one of 4 bucket classes."""
    if wlp < 0.250:
        return 0  # Very Poor
    elif wlp < 0.500:
        return 1  # Below Average
    elif wlp < 0.750:
        return 2  # Good
    else:
        return 3  # Elite


def bucket_name(bucket_id):
    return {0: "Very Poor", 1: "Below Average", 2: "Good", 3: "Elite"


def bucket_midpoint(bucket_id):
    """Return the midpoint W-L% for a given bucket (used as the numer
    return {0: 0.125, 1: 0.375, 2: 0.625, 3: 0.875}.get(bucket_id, 0.
```

```
_nb_data = _all_data.copy()
_nb_data["WL_bucket"] = _nb_data["W-L%"].apply(wl_bucket_label)

NB_FEATURE_COLS = ["SoS", "SRS", "OSRS", "DSRS"]

_nb_y_all = _nb_data["WL_bucket"].values
_nb_X_all = _nb_data[NB_FEATURE_COLS].values

from sklearn.model_selection import train_test_split

X_nb_train, _nb_X_temp, y_nb_train, _nb_y_temp = train_test_split(
    _nb_X_all, _nb_y_all, test_size=0.40, random_state=42
)
X_nb_val, X_nb_test, y_nb_val, y_nb_test = train_test_split(
    _nb_X_temp, _nb_y_temp, test_size=0.50, random_state=42
)

# keep actual W-L% values aligned for regression metrics later
_nb_wlp_all = _nb_data["W-L%"].values
_, _wlp_temp, _, _ = train_test_split(
    _nb_wlp_all, _nb_y_all, test_size=0.40, random_state=42
)
_, wlp_nb_test, _, _ = train_test_split(
    _wlp_temp, _nb_y_temp, test_size=0.50, random_state=42
)

print(f"NB features : {NB_FEATURE_COLS}")
print(f"Train (60%) : {len(X_nb_train)}")
print(f"Val   (20%) : {len(X_nb_val)}")
print(f"Test  (20%) : {len(X_nb_test)}")
```

```
NB features : ['SoS', 'SRS', 'OSRS', 'DSRS']
Train (60%) : 307
Val   (20%) : 102
Test  (20%) : 103
```

```
def train_model():
    scaler = StandardScaler()
    X_train_scaled = scaler.fit_transform(X_nb_train)
    model = GaussianNB()
    model.fit(X_train_scaled, y_nb_train)
    return model, scaler
```

```
def save_model(model, scaler):
    os.makedirs(NB_MODEL_FILEPATH, exist_ok=True)
    joblib.dump(model,  os.path.join(NB_MODEL_FILEPATH, "nb_wlp_bucke
    joblib.dump(scaler, os.path.join(NB_MODEL_FILEPATH, "scaler_wlp_bu


def evaluate_model(model, scaler):
    X_test_scaled = scaler.transform(X_nb_test)
    preds = model.predict(X_test_scaled)

    acc = accuracy_score(y_nb_test, preds)
    report = classification_report(
        y_nb_test, preds,
        target_names=["Very Poor", "Below Average", "Good", "Elite"],
        zero_division=0,
    )
    cm = confusion_matrix(y_nb_test, preds)

    predicted_wlp = np.array([bucket_midpoint(int(p)) for p in preds]
    actual_wlp = wlp_nb_test.astype(float)

    mse = mean_squared_error(actual_wlp, predicted_wlp)
    r2 = r2_score(actual_wlp, predicted_wlp)

    with np.errstate(divide='ignore', invalid='ignore'):
        pct_errors = np.where(
            actual_wlp != 0,
            np.abs((actual_wlp - predicted_wlp) / actual_wlp) * 100,
            np.nan,
        )
    mean_pct_err = np.nanmean(pct_errors)

    print("Naive Bayes - Test Set Performance")
    print(f"  Accuracy     : {acc:.4f}")
    print(f"  MSE (W-L%)   : {mse:.6f}")
    print(f"  R² (W-L%)    : {r2:.6f}")
    print(f"  Mean % Error : {mean_pct_err:.2f}%")
    print()
    print("Classification Report")
    print("-" * 40)
    print(report)
    print("Confusion Matrix")
    print("-" * 40)
    print(cm)
```

```
    pred_rows = []
    for i in range(len(y_nb_test)):
        actual_bucket = int(y_nb_test[i])
        pred_bucket = int(preds[i])
        actual_wl_val = float(actual_wlp[i])
        pred_wl_val = float(predicted_wlp[i])
        abs_err = abs(actual_wl_val - pred_wl_val)
        pct_err = float(pct_errors[i]) if not np.isnan(pct_errors[i])
        pred_rows.append({
            "Actual_WL_Percent"   : round(actual_wl_val, 3),
            "Predicted_WL_Percent": round(pred_wl_val, 3),
            "Absolute_Error"      : round(abs_err, 3),
            "Percent_Error"       : round(pct_err, 2) if pct_err is n
            "Actual_Bucket"       : bucket_name(actual_bucket),
            "Predicted_Bucket"    : bucket_name(pred_bucket),
            "Correct"             : int(actual_bucket == pred_bucket)
        })
    return pd.DataFrame(pred_rows)


def save_predictions(preds_df):
    os.makedirs(NB_PREDICTIONS_FILEPATH, exist_ok=True)
    preds_df.to_csv(
        os.path.join(NB_PREDICTIONS_FILEPATH, "predictions.csv"), ind
    )
```

```
nb_model, nb_scaler = train_model()
save_model(nb_model, nb_scaler)
print("Model saved.")

preds_df = evaluate_model(nb_model, nb_scaler)
save_predictions(preds_df)
print("\nPredictions saved.")
display(preds_df)
```

```
Model saved.
Naive Bayes - Test Set Performance
  Accuracy     : 0.7282
  MSE (W-L%)   : 0.012165
  R² (W-L%)    : 0.688459
  Mean % Error : 20.16%
```

```
Classification Report
----------------------------------------
               precision    recall  f1-score   support

    Very Poor       0.55      0.75      0.63         8
Below Average       0.71      0.73      0.72        37
         Good       0.75      0.73      0.74        41
        Elite       0.86      0.71      0.77        17

     accuracy                           0.73       103
    macro avg       0.72      0.73      0.72       103
 weighted avg       0.74      0.73      0.73       103

Confusion Matrix
----------------------------------------
[[ 6  2  0  0]
 [ 5 27  5  0]
 [ 0  9 30  2]
 [ 0  0  5 12]]

Predictions saved.
```

	Actual_WL_Percent	Predicted_WL_Percent	Absolute_Error	Percent_E
0	0.750	0.625	0.125	
1	0.313	0.375	0.062	
2	0.313	0.375	0.062	
3	0.375	0.375	0.000	
4	0.313	0.375	0.062	
...	...	...	...	
98	0.647	0.875	0.228	
99	0.588	0.375	0.213	
100	0.531	0.375	0.156	
101	0.750	0.625	0.125	
102	0.706	0.625	0.081	

103 rows × 7 columns

Next steps: Generate code with `preds_df` New interactive sheet